

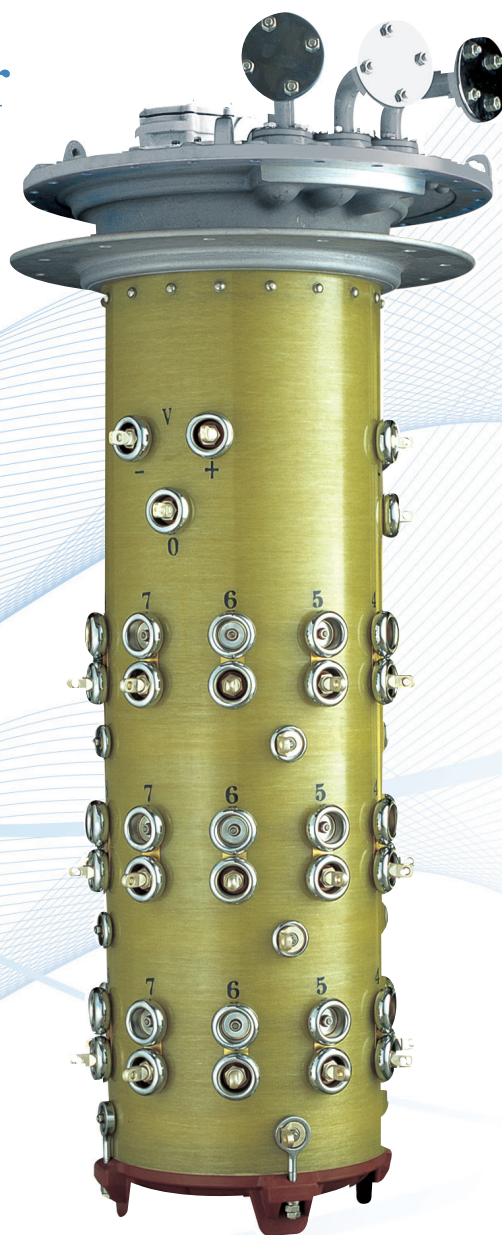


CV & SV

On-Load Tap-changer

Operating Instructions

HM 0.460.001-02.05/2021



IMPORTANT:

*Please read this manual carefully
before using the tap-changer and
save it for reference.*

Thank you for choosing Huaming products.

Before using, please read this instruction carefully. The product may have been modified since this document was published. We reserve the right to change the technical data, design and scope of supply without warning in advance. Generally, information provided with individual quotations and orders is binding. Huaming has no liability for any damage due to incorrect interpretation of this document.

The original document is written in Chinese.

Keep this manual at hand for future use and in case of any question, please contact Huaming.

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1.Foreword

This instruction manual is a comprehensive description of product transportation, installation, commissioning, test and maintenance. It is intended for trained and qualified staff related to transformer production, testing and quality assurance, and has reference value for transformer design and operation personnel.

1.1 Safety

Safety instructions are explained in this manual. They must be observed and followed to avoid relevant dangers.
Personal safety equipment must be worn during work to prevent potential risks.

	Protective clothing for protecting personnel from hazards in the work environment.
	Safety shoes for protecting personnel from falling sharp objects and slippery surfaces.
	Goggles for protecting eyes and facial area from impact, dust, liquid splash and high heat hazards.
	Hard hat for protecting head from injury.
	Hearing protection for protecting hearing from damage.
	Protective gloves for protecting hands from mechanical, chemical, and electrical hazards.

Table 1 Personal safety equipment list

1.2 Notation and conventions



DANGER!
The yellow triangle warning sign means life and health may be at risk if neglecting the instruction. Neglecting this warning can lead to serious or fatal injuries.



CAUTION!
The orange round warning symbol means products or device may be damaged if the instruction is not observed. Neglecting this warning can lead to product's or operator's damages and injuries.

**NOTE!**

The blue note symbol indicates important information that must be noted.

1.3 Safety precautions

Precaution	Action
Electric shock	Always unplug a device before inspection. Stay away from high-voltage equipment and wear protective clothing.
Transformer oil	Collect used transformer oil in dedicated drums. Keep it away from flames or sparks. Do not touch it which may cause damages to health.
Fire	In case of fire, extinguish it with powder, foam or carbon dioxide. Do not use water.
Waste	Wastes are dangerous to the environment or human health. Clean them up in time and use different methods according to the wastes' categories.

Table 2 Safety precaution

2. General

CV/SV on-load tap-changer, used to regulate voltages, is suitable for power or industrial transformers with rated voltages of up to 110kV, a maximum rated current of three-phase 500A, single-phase 700A, and a frequency of 50Hz or 60Hz.

CV/SV is selector switch type OLTC which combines the duties of a diverter switch and a tap selector in one compartment.

They are installed on the transformer tank cover via the tap-changer head flange, and is connected to the CMA7 or SHM-D motor drive unit through the worm gear reducer, horizontal drive shaft, bevel gear box, and vertical drive shaft, so as to achieve the purpose of tap change.

**NOTE!**

Special design is available. Please contact Huaming for further information.

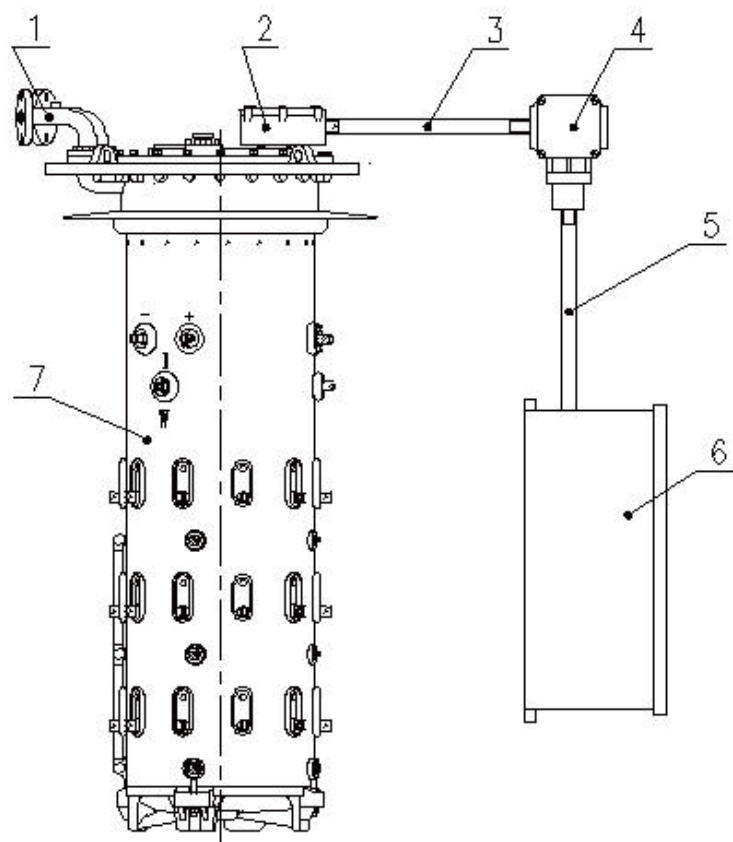


Figure 1 Overview of tap-changer

1. Pipe bend	2. Upper gear unit
3. Horizontal drive shaft	4. Bevel gearbox
5. Vertical drive shaft	6. Motor drive unit
7. Oil compartment	

Table 3 List of tap-changer components

3. Technical data

CV/SV tap-changer complies with IEC 60214-1: 2014 standard.

3.1 Table of technical data

Description		CVIII 350Y	CVIII 350D	CVI 350	SVIII 500Y	SVIII 500D	CVI 700
Max. rated through current (A)		350	350	350	500	500	700
Rated frequency (Hz)		50 or 60					
Connection		3-phase Y-connection for neutral point only 3-phase D-connection for any connection except neutral point Single-phase for any selectable winding connection					
Max. rated step voltage (V)	10 contacts	1500			1500	1500	
	12 contacts	1400			1400	1400	
	14 contacts	1000			--	1000	
Rated step capacity (kVA)	10 contacts	525			400-525*	660	
	12 contacts	420			325-420*	520	
	14 contacts	350			--	450	
Short-circuit current test (kA)	Thermal (3s)	5			7	10	
	Dynamic (peak)	12.5			17.5	25	
Max. operating positions		See Figure 4					
Insulation to ground	The highest voltage for equipment (kV)	40.5			72.5		
	Rated separate source AC withstand voltage (kV/50Hz, 1min)	85			140		
	Rated lightning impulse withstand voltage (kV,1.2/50μs)	225			350		
Mechanical life		Not less than 800,000 operations					
Electrical life		Not less than 200,000 operations					
Oil compartment of diverter switch	Service pressure	0.03MPa					
	Leakage test	No leakage under 0.1 MPa for 24 hours					
	Over pressure protection	Rupture disc bursts at 300 ± 20% KPa					
	Protective relay	Set oil flow speed at 1.0m/s ± 10%					
Motor drive unit		CMA7 or SHM-D					

*Note: The max. through current should decrease to 350A and 300A when the switching capacities are 525 kVA and 420 kVA.

Table 4 CV/SV technical data

3.2 Operation conditions

- 1). The operating temperature of the tap-changer in oil is $\geq -25^{\circ}\text{C}$ and $\leq 105^{\circ}\text{C}$.
- 2). The storage environment temperature of the tap-changer is $\geq -25^{\circ}\text{C}$ and $\leq 40^{\circ}\text{C}$, and the humidity should be not higher than 85%. If beyond this standard, please mark your requirements before ordering. The tap-changer material and accessories will be specially processed to meet the special requirements of the environment.
- 3). After being mounted to the transformer, the non-perpendicular degree between the tap-changer and the ground should be less than 2° .
- 4). No sever dust and other explosive or corrosive gases on the installation site.

**CAUTION!**

Do not store the tap-changer or motor drive unit in a place with high humidity ($\geq 85\%$). This will cause condensation.

**NOTE!**

Special design is available to meet particular environment conditions. Please contact Huaming for further information.

4. Type designation

4.1 Type designation

CV/SV tap-changers have a variety of specifications due to different combinations of the number of phases, the maximum rated through current, the highest voltage of the equipment, the insulation class of the selector and the connection mode. Therefore, the above-mentioned performance parameters should be clearly reflected in the model of the tap-changer, and the detailed description of each code is shown in Figure 2.

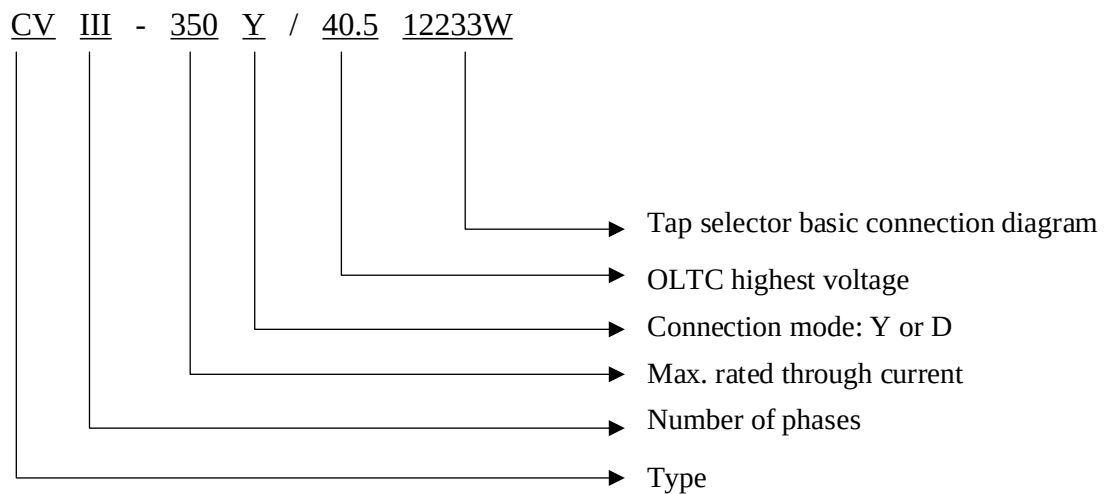


Figure 2 CM2 model designation

4.1.1 Basic connection

According to the transformer voltage regulation and winding connection mode, the tap selector has a variety of different specifications. It is composed of the number of tap selector pitch, the number of operating positions and the form of the changeover selector. The designation of the tap selector model is shown in Figure 3.

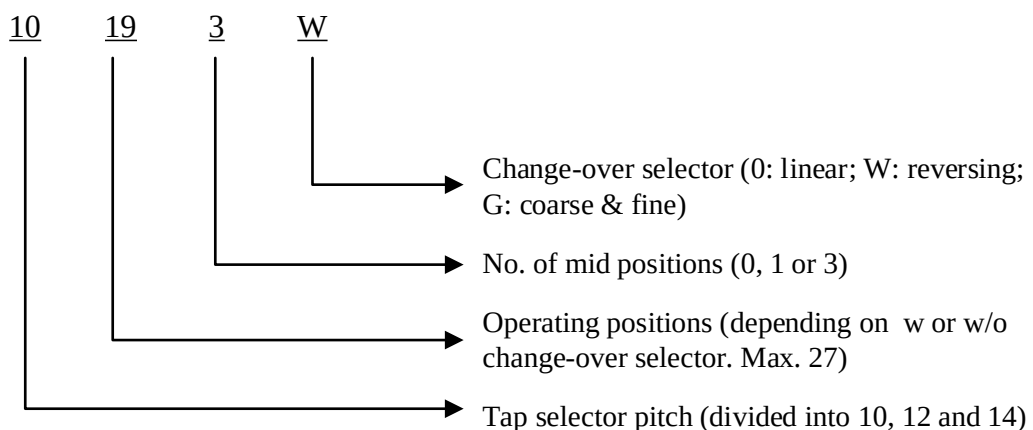


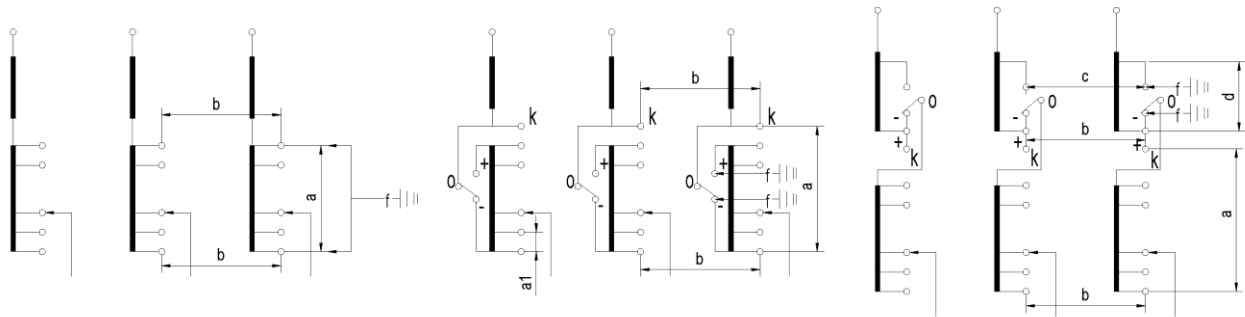
Figure 3 Tap selector basic connection

For example:

- 1). Linear regulation: It is represented by 5 digits. For example, 14140 means that the contact position of each phase is 14, the number of working positions is 14, and the number of intermediate positions is 0.
- 2). Reversing regulation: Add a letter W after 5 digits, such as 14131W, which means that the contact position of each phase is 14, the number of working positions is 13. The number of intermediate positions is reversing adjustment.
- 3). Coarse and fine regulation: 5 digits followed by a letter G, such as 14131G, indicating the contact position of each phase 14, the number of working positions is 13. The number of intermediate positions is coarse and fine adjustment.

4.2 Internal insulation level

The tap-changer internal insulation level is shown in Table 5.



unit(kV)

Insulation distance		Tolerating voltage	Type of tap-changer		
			CVIII350Y	CVIII350D	CVI350
			SVIII500Y	SVIII500D	CVI700
a	10 contacts	1.2kV/50μs	200		
		kV50Hz 1min	50		
	12 contacts	1.2kV/50μs	180		
		kV50Hz 1min	50		
	14 contacts	1.2kV/50μs	170		
		kV50Hz 1min	50		
b	40.5kV	1.2kV/50μs	200	225	--
		kV50Hz 1min	70	85	--
	72.5kV	1.2kV/50μs	200	350	--
		kV50Hz 1min	70	140	--
c	40.5kV	1.2kV/50μs	350	350	--
		kV50Hz 1min	140	140	--
	72.5kV	1.2kV/50μs	350	350	--
		kV50Hz 1min	140	140	--
d		1.2kV/50μs	200		
		kV50Hz 1min	53		
f	40.5kV	1.2kV/50μs	225		
		kV50Hz 1min	85		
	72.5kV	1.2kV/50μs	350		
		kV50Hz 1min	140		

Table 5 Insulation level of Tap selector

4.3 Tap selector wiring diagram

The different transformer voltage winding tapping methods correspond to different specifications of tap selector's basic wiring diagrams. Figure 4 shows the common basic wiring diagrams. Special designs can be made to meet customers' requirements.

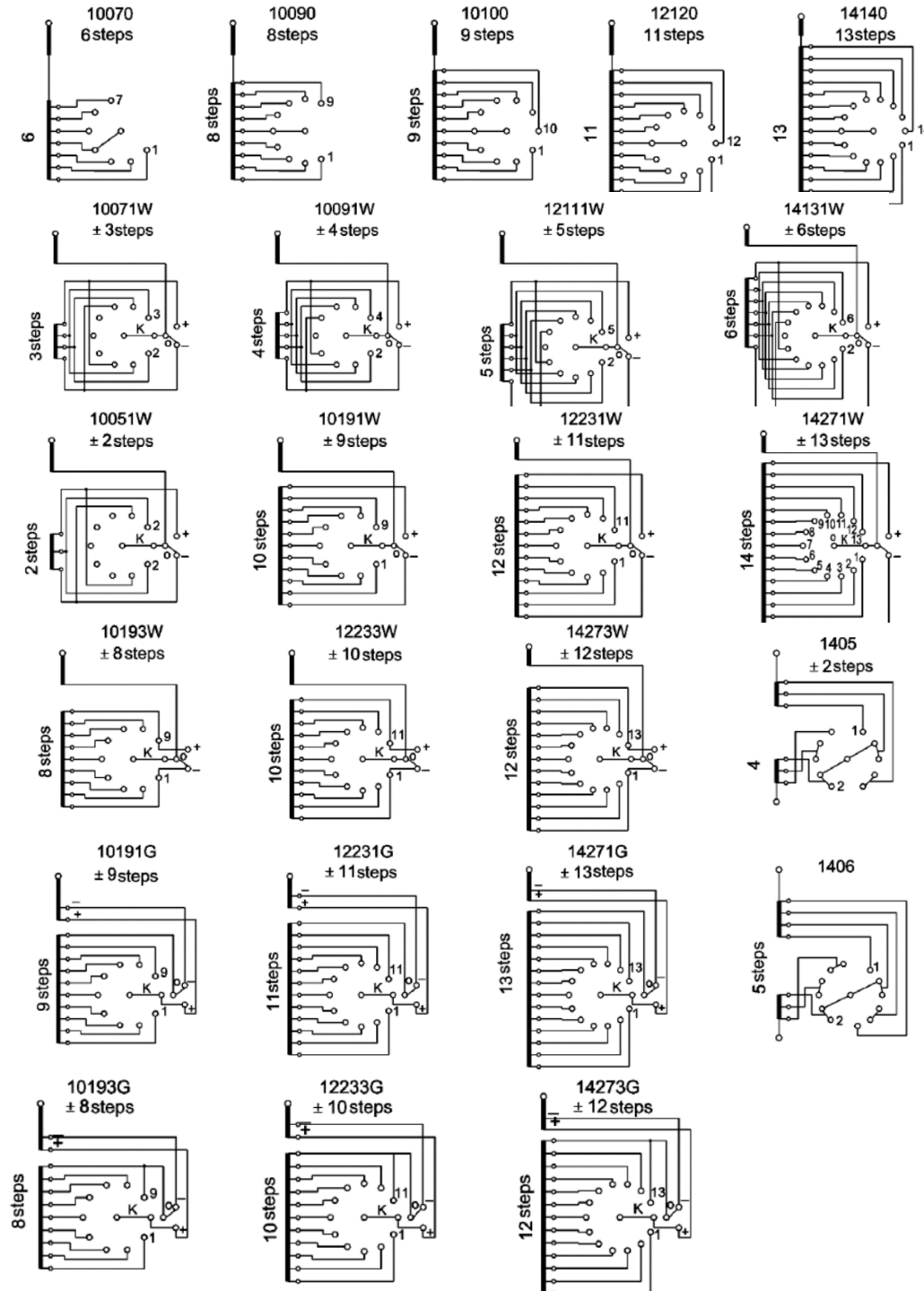


Figure 4 Basic connection diagram of Tap selector

5. Packing, transportation & storage

5.1 Supplied components

The complete set of tap-changer contains below parts (Figure 5):

- Tap-changer
- Motor drive unit (MDU), controller and connecting cables, transmission shafts, couplings, bevel gear boxes and other transmission accessories
- Protection relay and accessories
- Special tools and accessories for tap-changer assembly
- Customer-required special accessories if any.

Please refer to the corresponding packing list delivered with the tap-changer for exact supplied components.



Figure 5 Set of tap-changer

5.2 Transportation & acceptance

5.2.1 Transportation

- Packings is suitable for all common kinds of transportations. The boxes of tap selector and MDU are allowed to be appropriately stacked during storage, and the load-bearing capacity of the box's top surface shall not exceed 500kg/m^2 .
- Boxes should be placed reasonably according to the position of the gravity center. When lifting it up, the markings on the box must be observed (Figure 6).



Figure 6 Example of markings on the box

**DANGER!**

When lifting and storing the boxes, people are not allowed to stand under or near the boxes in case the boxes drop. All boxes must be stored properly and safely.

**CAUTION!**

Choose appropriate forklift and crane according to the dimension and gross weight of the boxes.

5.2.2 Goods receiving

Each batch of delivered goods must be checked by the consignee before accepted (signing acceptance confirmation). Following items must be checked by the consignee carefully:

- When receiving the goods, check the quantity of components received according to the packing list.
- Check whether the packing is intact. If it's found to be damaged (if possible, take pictures of the box and damaged goods) or the information of the goods does not match the packing list, consignee should refuse to sign for acceptance, and immediately report the problem to the shipping company and Huaming.
- The above steps are also applicable to corrosion of the goods due to moisture immersion (rain, snow, condensed water). The components must be stored in a dry place before installation.
- The on-load tap-changer must be stored in a sealed packing box and opened before installation only.

**NOTE!**

When receiving the goods, please check carefully the packing and the supplied components according to the packing list. In case anything is broken or missing, please contact Huaming in time.

**NOTE!**

The transportation and lifting of the box must be operated by professionals. The carrying capacity of transportation vehicles and cranes should be more than 1000kg.

**CAUTION!**

The packing box should be firmly fixed during transportation and lifting, and be handled with care to avoid possible damage to the product due to vibration, impact, falling, tipping, bumping, shaking, etc.

**CAUTION!**

If the box drops or is severely impacted, the possibility of damages must be considered. Comprehensive inspection of the product should be carried out, and if necessary, inspection(repair) shall be done by the Huaming's professional and technical personnel.

5.2.1 Acceptance

Unpack the box and look for damage during transportation.

- Transport the box to the installation site.
- Check the equipment and its accessories according to the packing list.

**CAUTION!**

During unpacking and inspection, avoid damaging the original package.

**NOTE!**

If the received goods and its accessories do not match the list, contact Huaming in time.

5.3 Storage

When the goods are stored continuously for more than 1 year, the equipment should be inspected thoroughly before installation.

Specially packed goods with sealed packing can be stored outdoors if they meet the following conditions.

- The storage environment temperature should not be $\geq -25^{\circ}\text{C}$ and $\leq 40^{\circ}\text{C}$.
- The tap-changer must not be stored in an environment with flammable, explosive and corrosive gases.
- It must be protected from moisture, dust, rodents and ants.
- Regularly check whether the stored equipment is abnormal
- If the tap-changer is stored for a long time, the desiccant must be replaced regularly, and re-packed with sealed packing.

**CAUTION!**

After long-term storage, the tap-changer and its accessories must be carefully inspected before installation.

6. Mounting of tap-changer on the transformer

6.1 Transformer mounting flange for tap-changer installation

A mounting flange is required to fix the tap-changer head on the transformer cover. The structure of the flange should be designed according to the shape of the tap-changer head's sealing surface (Figure 7). Double-thread screw (M12, maximum length=45mm) should be accurately positioned.

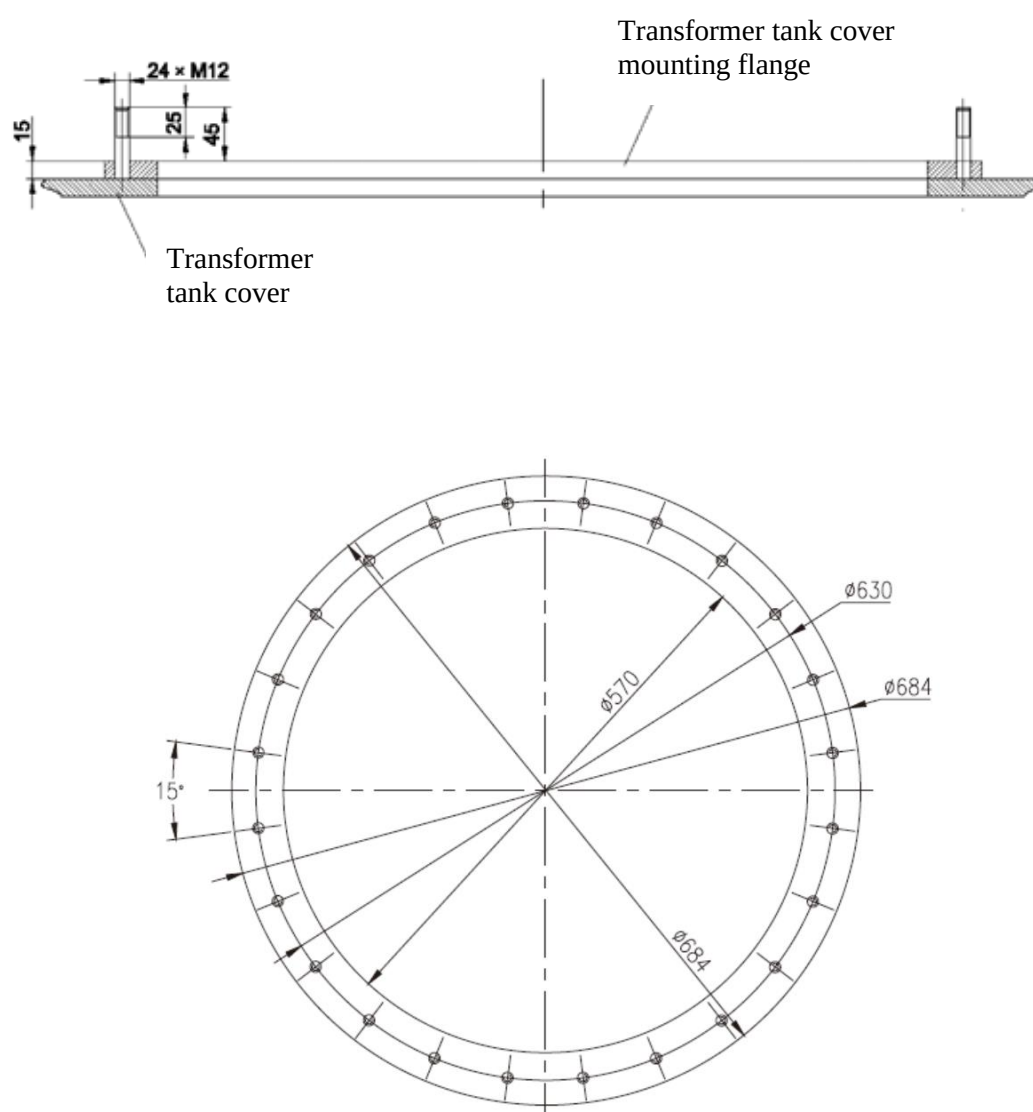


Figure 7 Transformer mounting flange

6.2 Mounting of tap-changer on standard type transformer

- 1) Clean the bottom sealing face of the tap-changer head flange as well as the surface of transformer mounting flange, and place a sealing gasket on it (Figure 8).
- 2) Lift the assembled tap-changer to be above the transformer's mounting flange. Carefully place it into the transformer tank through the mounting hole on the transformer cover (Figure 9).

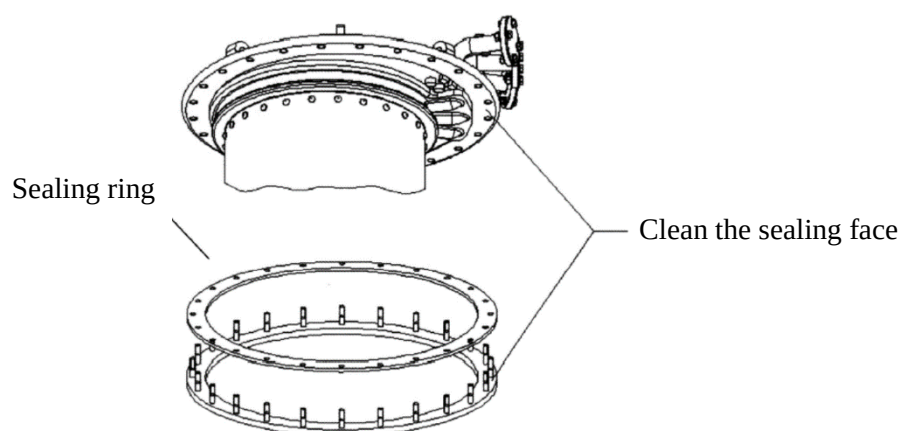


Figure 8 Sealing face

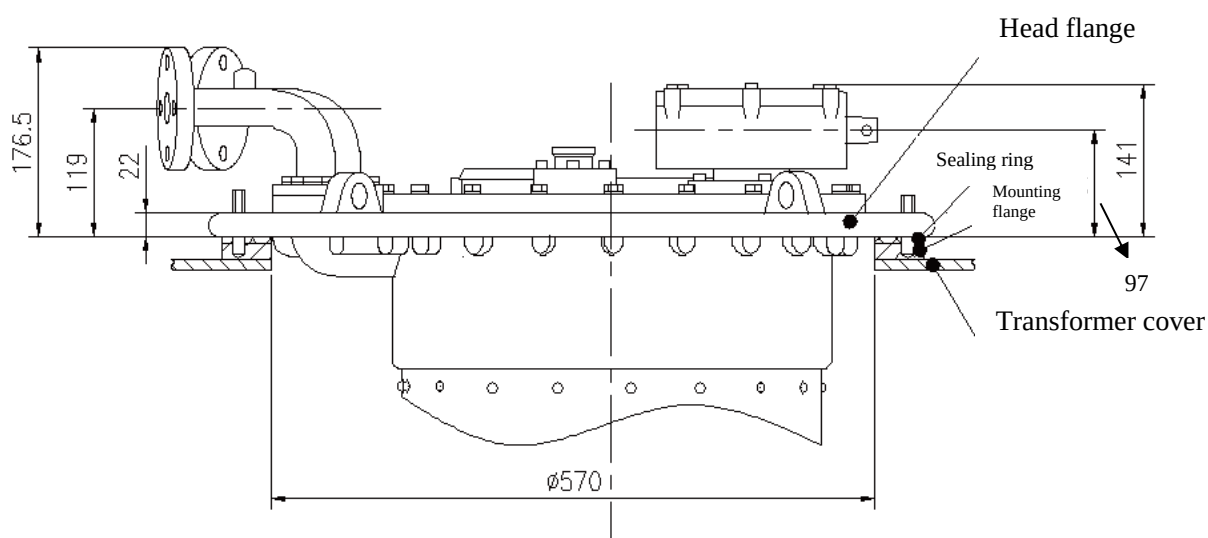


Figure 9 Mounting hole

- 3) Check whether the tap-changer head's mounting position and angle meet the design requirements. If yes, fix the tap-changer head flange to the transformer's mounting flange, and fasten 24 pieces of nuts with a torque of 100 to 110 N.m.

**CAUTION!**

Make sure not to damage the tap-changer's wiring terminal and tie-in resistors, if any.

**DANGER!**

If heavy components are not placed on a flat surface, they may tilt and cause injuries! Therefore, they must be put on a horizontally flat surface.

6.3 Mounting of tap-changer on bell-type transformer



NOTE

- 1) Make sure small parts do not fall into OLTC oil compartment.
- 2) During disassemble and reassemble, all small parts must be complete. Make sure to check carefully.

6.3.1 Disassembly of tap-changer head cover

- 1) Make sure the tap-changer is at the calibration position.
 - 2) Unscrew the 24 pieces of M10 bolts on the tap-changer head cover (with spring washer, 17mm wrench).
 - 3) Remove the tap-changer head cover (Figure 10).
- During disassembly and other operations, avoid damaging the head cover and the sealing surface (O-ring) (Figure 11).

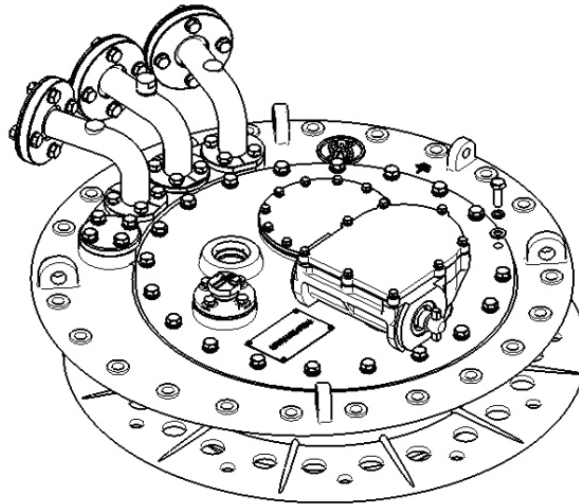


Figure 10 Tap-changer head cover

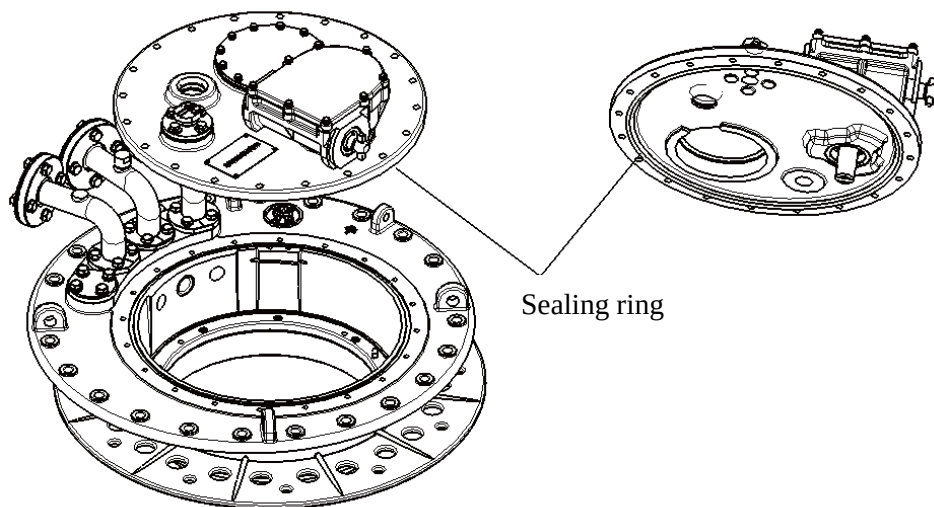


Figure 11 Tap-changer sealing ring

6.3.2 Disassembly of spring energy system

The bottom plate of the spring energy system is fixed with 5 pieces of M8 bolts (Figure 12). (Remember the spring energy system's marked place.)

- 1) Use 20 pieces of M5 bolts to pull out the spring energy system's fixing pins. Remember the pins' places for future re-assembly (Figure 13).
- 2) Unscrew and remove oil suction pipe R's 3/4" connector (27mm and 39mm wrench). Pay attention to the sealing gasket.
- 3) Turn pipe R clockwise for 45°.
- 4) Loosen 5 pieces of M8x20 screws (13mm wrench) and take care of the spring gasket.
- 5) Pull out the spring energy system in upward direction (Figure 14).

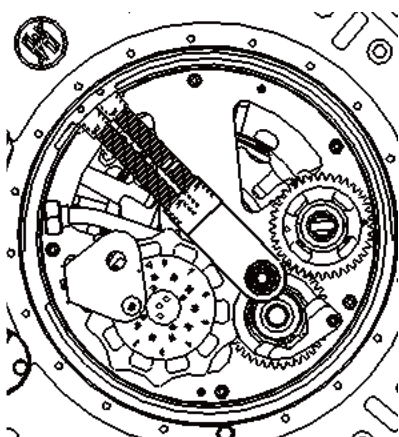


Figure 12 Spring energy system

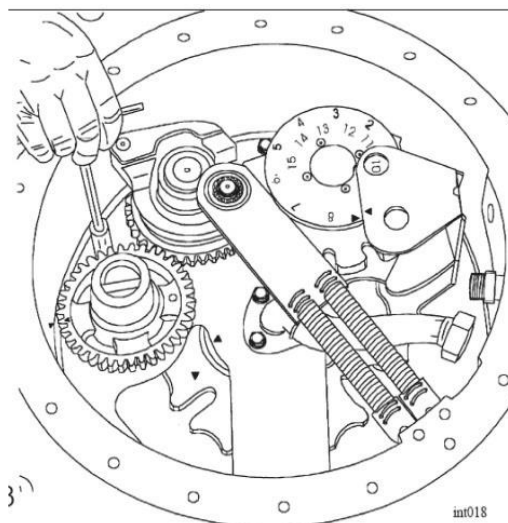


Figure 13 Pull out the positioning pin

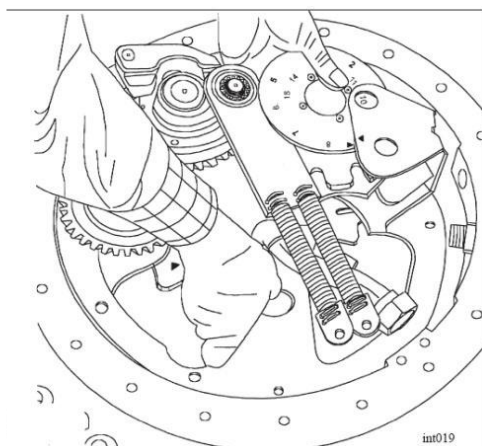


Figure 14 Pull out the spring energy system

**NOTE!**

Keep all removed parts in a safe place for re-assembly later.

6.3.3 Lifting out the tap-changer switch insert

- 1) Use a screwdriver to pry out the oil suction pipe, and carefully take it out from main shaft
- 2) Lift out the switch insert out of the oil compartment (Figure 15 & 16)) by using tap-changer lifting device (Appendix.4)

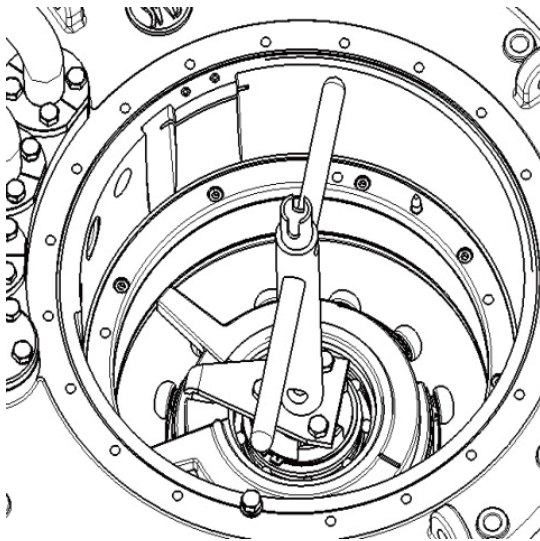


Figure 15 Install lifting device

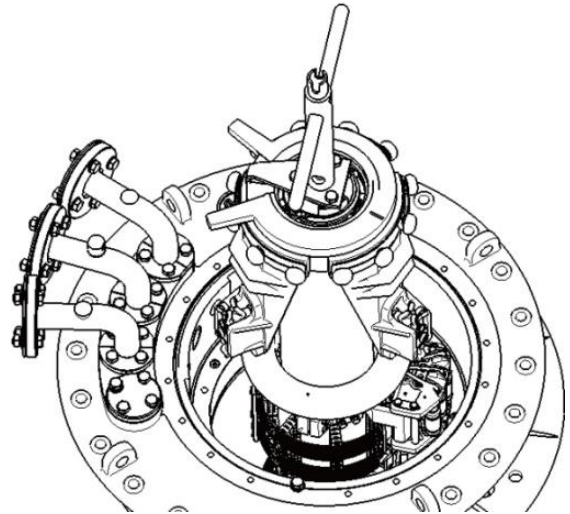


Figure 16 Lifting out the main shaft

To avoid bumping and scratching the moving contacts, it is suggested to lift out the insert in two steps. First, lift up for around 60mm. Then turn about 10° to stagger the insert's moving contacts and the oil compartment's fixed contacts. After that, lift up and take out the switch insert.

Re-assembly is in opposite sequence. First push the switch insert hard all the way to the bottom. Make sure it is firmly connected to the bottom.

**CAUTION!**

Use the special lifting device. Otherwise, the switch insert may be damaged.

**CAUTION!**

During lifting process, be careful not to let the insert bumping the oil compartment, which may damage both of them.

6.3.4 Disassembly of tap-changer head flange and main body

Loosen the fixing screws of the tap-changer head flange (9 pieces of M8 socket head screws with spring gasket, and store them in a safe for re-assembly later).

**CAUTION!**

Pay special attention to avoid any parts dropping into the tap-changer oil compartment. Otherwise, the tap-changer may get stuck during operation and both the changer and transformer will be damaged. Therefore, all small parts must be counted to make sure none of them are missing during disassembly and re-assembly.

**CAUTION!**

When lifting the head flange from the tap-changer support flange, keep an eye on the O ring between these two flanges (Figure 17).

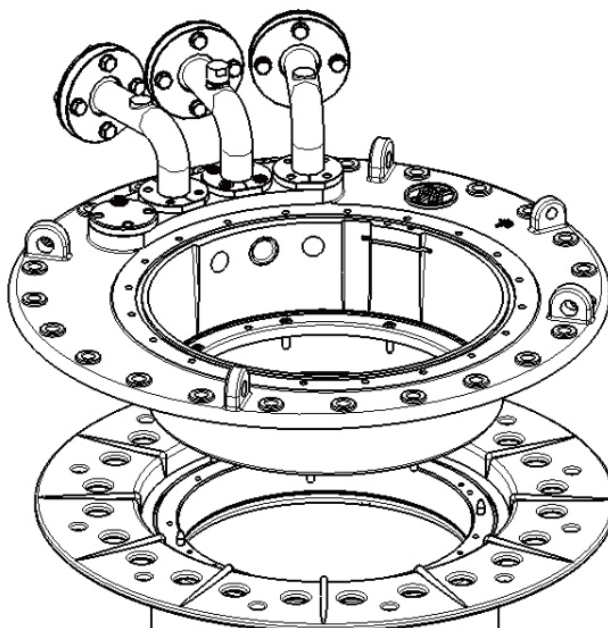


Figure 17 Lifting the tap-changer head flange

6.3.5 Put oil compartment on the bracket

Use a special lifting tool to lift the tap-changer oil compartment onto the bracket. Check the install position and then fix it on the bracket (Figure 18).

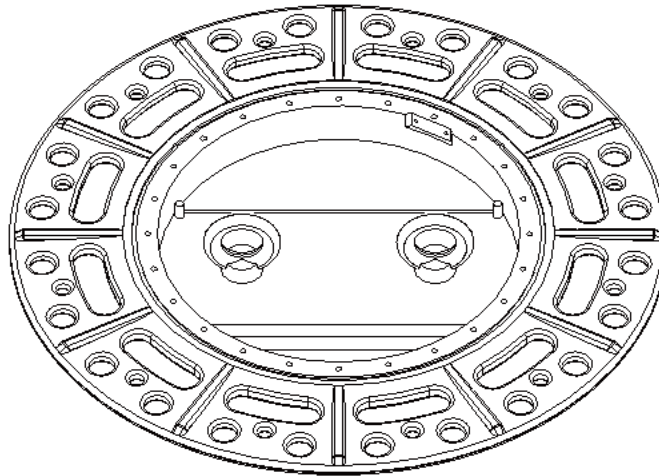


Figure 18 Lifting the tap-changer head flange



CAUTION!

Use the professional lifting tools during installation. Otherwise, the OLTC oil compartment might be damaged.

Connect the regulating winding taps to tap-changer winding terminals according to the markings on the oil compartment (Refer to next section for more information).



CAUTION!

The connected cables should not have any pulling force on the OLTC. There should have enough space for wires, so the OLTC can be raised to the final installation location after closing the bell-type transformer cover.

6.3.6 Assembly of bell-type transformer tank cover

- 1) Before assembling bell-type transformer tank cover, clean the tank cover and tap-changer supporting flange's sealing surface.
- 2) Lift the tank cover to be above the transformer tank and slowly lower it.
- 3) Before mounting tap-changer head flange, clean the sealing surfaces of the mounting flange and head flange. Place a sealing ring on the mounting flange, and then place the tap-changer head flange on the mounting flange.
- 4) There should be space between the head flange and mounting flange, which, depending on the final height, is around 5 to 20mm. Check the installation of the tap-changer head with the help of two rods, because two rods can make sure the installation position of OLTC head and supporting flanges is correct.
- 5) Fix the tap-changer head flange on the transformer mounting flange (24 pieces of M10 bolts, 17mm wrench, max. torque 14 N.m.).
- 6) Use the special lifting tool to slightly lift up the tap-changer oil compartment, and fasten the tap-changer head flange to the supporting flange (9 pieces of M8 bolts, max. torque 14 N.m.).

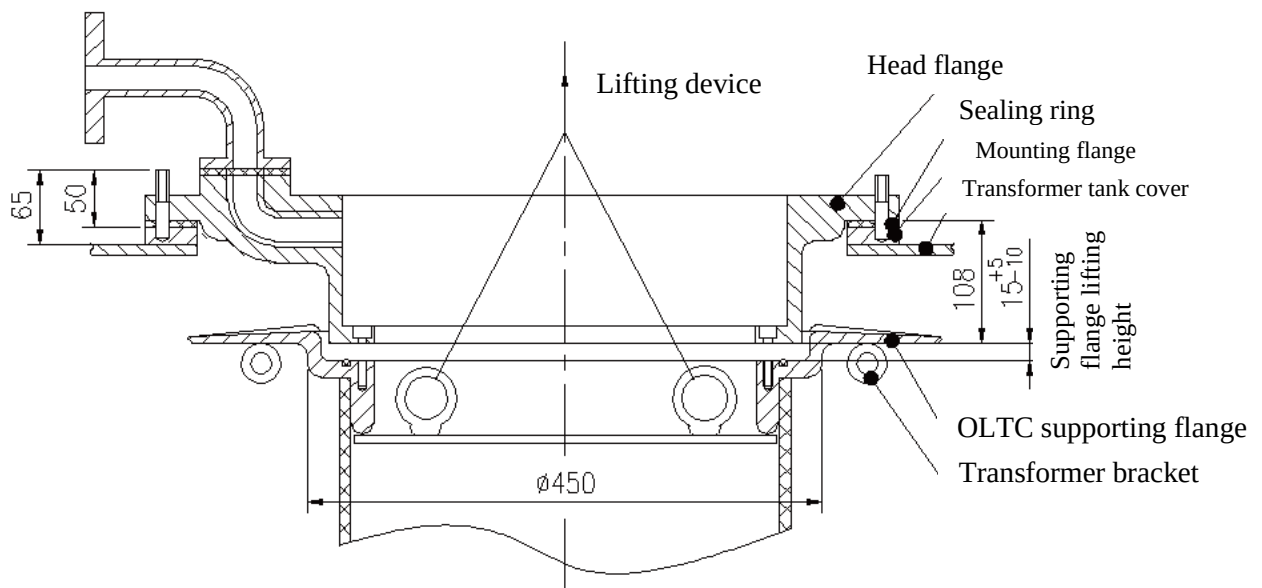


Figure 19 Bell-type OLTC installation

6.3.7 Re-assembly of oil suction pipe, switch insert and spring energy system & fixation of OLTC head cover

- 1) Re-assembly of oil suction pipe, switch insert and spring energy system please refer to their removal procedure, in reverse order.
- 2) Fixation of the tap-changer head cover: Before closing the head cover, check the sealing gasket. Use 20 pieces of M10x35 bolts (17mm wrench, 35 N.m.).

**CAUTION!**

During re-assembly, the tap-changer must be at the set position.

6.4 Connection between tap of voltage regulating winding and terminal of tap-changer

The tap of the voltage regulating winding must be connected according to the wiring schematic diagram.

**NOTE!**

Mechanical stress may damage the OLTC.

**CAUTION!**

The wiring of all tapping leads connected to the tap-changer must be reliable and tight. The assembly of tap lead shall be such that the lead connected to the tap selector does not pull the terminal.

**CAUTION!**

The distance between the cables and the OLTC surface must be no smaller than 50mm.

The diameter of the tap-changer fixed contact installation hole is 11mm, and the connecting leads are fastened by M10x35 bolts. Connecting between current led-out contact forms the neutral point and shall not be dismantled (Figure 20).

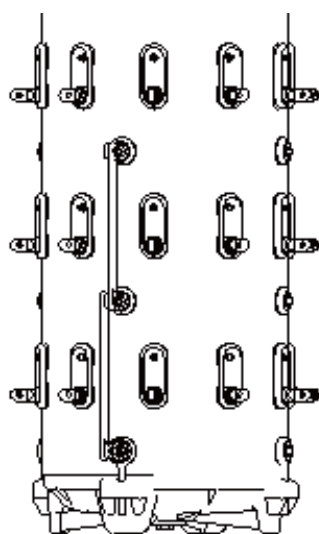


Figure 20 Neutral point led-out terminals

7. Transformer ratio test and transformer DC resistance measurement

**CAUTION!**

When without oil and before drying, OLTC shall not operate more than 10 cycles.

**CAUTION!**

Incomplete tap change operation can cause damage to the OLTC!

**NOTE!**

It is suggested that the ratio test and DC resistance measurement should be carried out before drying.

7.1 Transformer ratio test

When turning the drive shaft of the tap-changer upper gear unit (Figure 21), a short pipe with an inner diameter of 25 mm (a connecting bolt with a diameter of 12 mm has been screwed into the pipe) can be used, and then a hand wheel or hand crank is also needed.

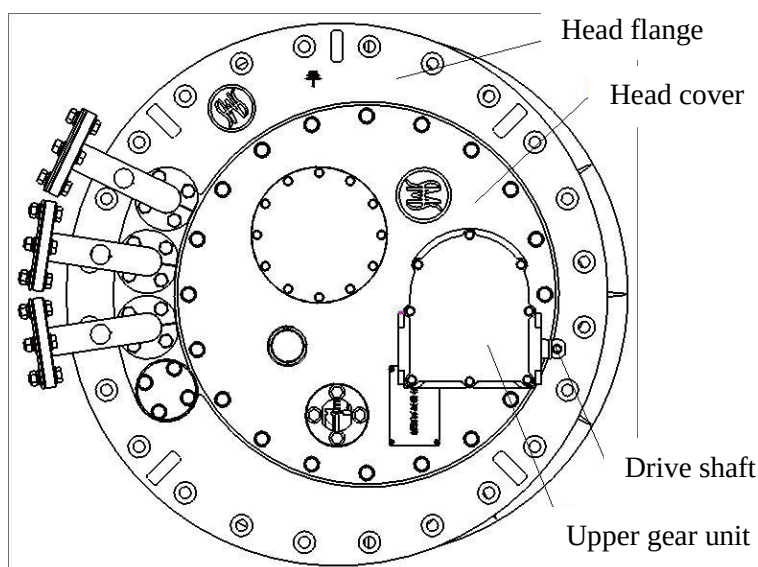


Figure 21 View of the OLTC head

For two or three sets of linkage OLTC group, all tap-changer heads should be connected with horizontal shaft. For a tap change, the sound of the diverter switch action can be heard clearly. During operation, each operation position should be observed at any time through the observation window on the tap-changer head (Figure 22), so as to avoid exceeding the end position of the tap-changer.

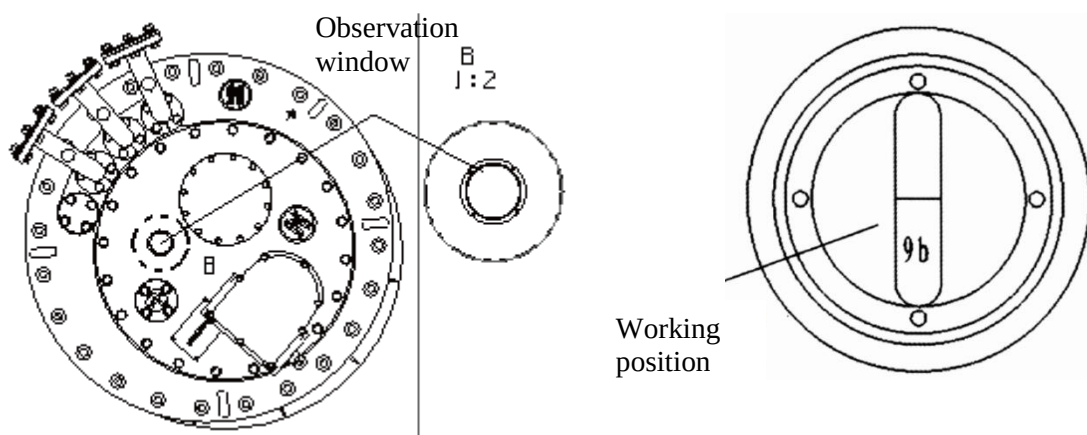


Figure 22 Observation window



CAUTION!

1. The tap-changer can only be operated through the drive shaft of the gear box.
2. In the case of no oil, too much operation will cause damage to the tap-changer!
3. After the transformer ratio test and DC resistance measurement, the OLTC must be adjusted to the set position.

7.2 Transformer DC resistance measurement

It should be noted that the tap-changer is operated without oil when measuring the DC resistance of transformer, and the operation of tap-changer should be reduced as much as possible.

After the transformer transformation ratio test and DC resistance measurement, the tap-changer must be adjusted back to the calibration position.

8. Drying process and oil filling

8.1 Drying process

To ensure the insulation property of tap-changer, necessary drying must be taken as per below description (vacuum drying or vapor phase drying).

8.1.1 Vacuum drying

**CAUTION!**

OLTC top cover, transmission gear box and OLTC accessories are not allowed to be vacuum dried together with the transformer in the drying oven. Otherwise, they will be damaged.

**CAUTION!**

Before drying, OLTC top cover, transmission gearbox, MDU, protective relay, pressure relief valve, bevel gearbox, online oil filter and other accessories must be removed and cannot be dried.

8.1.1.1 Drying in the oven room

- 1) The tap-changer is heated up at a rate of $10^{\circ}\text{C}/\text{hour}$ in the air under atmospheric pressure, till the temperature reaches highest $105^{\circ}\text{C} \pm 5^{\circ}\text{C}$.
- 2) Drying is carried out in the circulating hot air for 8 to 10 hours and highest temperature of tap-changer is $105^{\circ}\text{C} \pm 5^{\circ}\text{C}$.
- 3) Vacuum drying lasts 20 hours, temperature of tap-changer is 105°C till 125°C . Maximum residual pressure 133Pa.

8.1.1.2 Drying in the transformer oil tank

**CAUTION!**

During drying, the OLTC head cover is sealed all the time. Therefore, the OLTC must be connected to a pipe for vacuum process.

**NOTE!**

OLTC head cover can withstand the vacuum pressure.

By-pass pipe (nominal diameter 25mm) can be connected to the flange connection position between flanges E2 and Q.

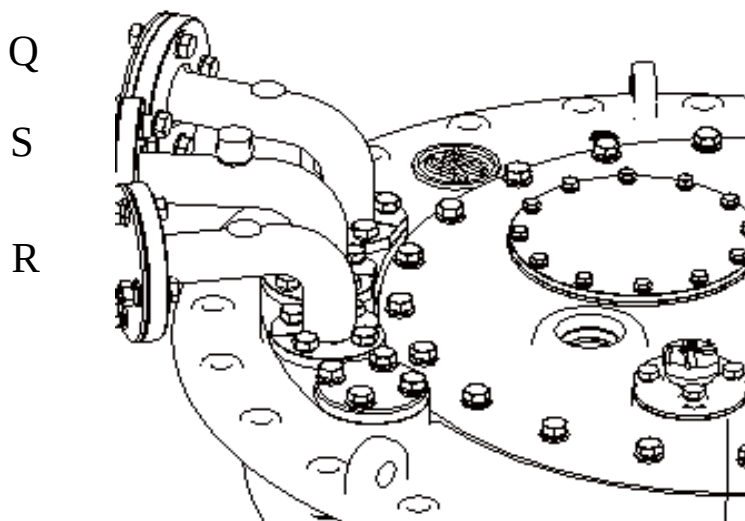


Figure 23 Pipe position

8.1.2 Vapor phase drying



CAUTION!

OLTC top cover, transmission gear box and OLTC accessories are not allowed to be vacuum dried together with the transformer in the drying oven. Otherwise, they will be damaged.

Before drying process, tap-changer top cover, transmission gear box, motor drive unit, protective relay, pressure monitoring device, pressure relief device, bevel gearbox, temperature sensor, online oil filter and so on must be removed and placed outside the drying oven. In terms of vapor phase drying, be sure to open the oil drain stopper at the bottom of the oil compartment before drying, so that the condensate can be discharged from the oil compartment.

8.1.2.1 Vapor phase drying inside the vacuum tank

1) Unscrew the oil drain stopper at the bottom of the oil compartment (socket head screw, 24mm wrench) (Figure 24).

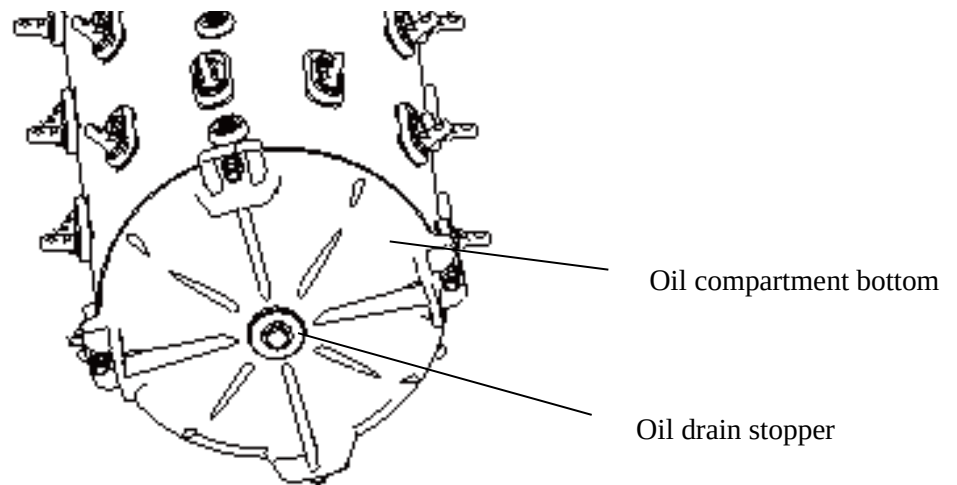


Figure 24 Unscrew the oil drain stopper

- 2) Unscrew the 24 pieces of M10 bolts on the tap-changer top cover (with washers).
- 3) Remove the tap-changer top cover and store it outside the vacuum oven.
- 4) Remove all accessories except the tap-changer body, including motor drive unit, relay, pressure monitoring device, PRD, gearbox, temperature sensor, online oil filter, etc., and put them outside the drying oven.
- 5) Pass in kerosene vapor around 90°C and keep this temperature constant for 3 to 4 hours.
- 6) Increase the kerosene gas temperature at a rate of 10°C/hour till the specified final temperature, but maximum temperature of the tap-changer cannot exceed 125°C.
- 7) In vacuum drying, the temperature of the tap-changer is 105°C to a maximum of 125°C. The highest residual pressure is 133Pa. The drying duration is the same as the transformer, but at least 50 hours.
- 8) Close the oil drain bolt (torque 18~20N.m.) after drying process.

8.1.2.2 Vapor phase drying inside the transformer tank

**CAUTION!**

Oil leakage will lead to OLTC and transformer damages!

**CAUTION!**

Tap-changers pipes R and Q (refer to Figure 11 and Chapter 8.1.1.2 for their positions) must be connected to by-pass pipes with an inner diameter of at least 25mm. Pass in Pass in kerosene vapor.

**NOTE!**

Damaged OLTC O ring will cause oil leakage from oil compartment which will damage the OLTC.

- 1) Pass in kerosene vapor around 90°C and keep this temperature constant for 3 to 4 hours.
- 2) Increase the kerosene gas temperature at a rate of 10°C/hour till the specified final temperature, but maximum temperature of the tap-changer cannot exceed 125°C.
- 3) In vacuum drying, the temperature of the tap-changer is 105°C to a maximum of 125°C. The highest residual pressure is 133Pa. The drying duration is the same as the transformer, but at least 50 hours.
- 4) Close the oil drain bolt (torque 18~20N.m.) after drying process.

8.2 Oil filling



CAUTION!

When filling the oil, do not step on the rupture disk (Figure 25). Do not fill the oil through the rupture disk's hole (Figure 26).

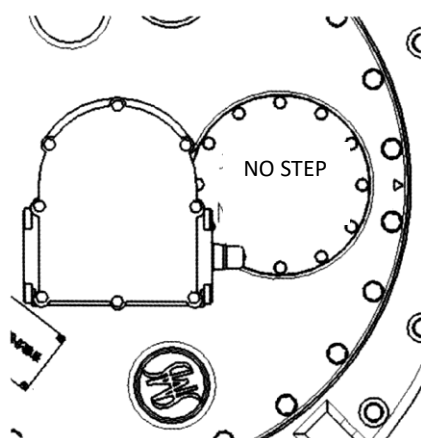


Figure 25 Do not step on the rupture disk

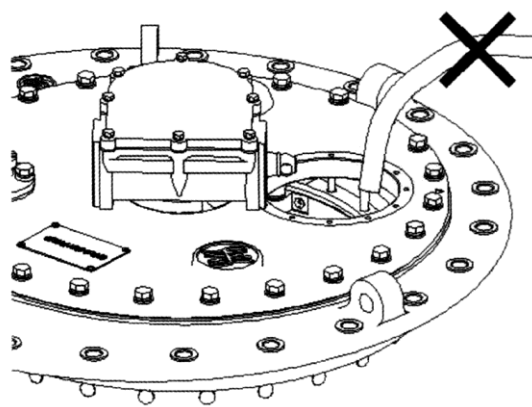


Figure 26 Do not fill the oil through the rupture disk's hole!

After drying, in order to prevent too much moisture from entering the oil compartment, the oil compartment (diverter switch insert already installed) should be refilled with oil as soon as possible. In order to meet the standard for oil filling of oil compartment and its conservator, use only new transformer insulating fluid in accordance with IEC 60296 for mineral oil, IEC 61099 for synthetic ester oil and IEC 62770 for natural ester oil.



NOTE!

Oil filled in the tap-changer's oil compartment and conservator must meet IEC 60296 and IEC 60422 standards.

Both the on-load tap-changer and the transformer should be filled with new oil under vacuum at the same time.

- 1) In order to vacuumize the tap changer and the transformer at the same time, connect a by-pass pipe and the appropriate valves between pipe joints E2 and Q.
- 2) When filling oil, use the pipe joint S or Q on the tap-changer head.
- 3) Do not dismantle the flexible by-pass pipes before shutting the connected valve
- 4) For a better and precise evaluation of insulating fluid, measure the breakdown voltage (as per IEC 60156) and water content (as per IEC 60814) at a sample of $20^{\circ}\text{C} \pm 5^{\circ}\text{C}$ (as per IEC 60156) and do not forget to record temperature of the sample.
- 5) Check breakdown voltage and water content values specified at table-7.



CAUTION!

1. Oil compartment without oil or with too little oil may damage the tap-changer.
2. After drying the tap-changer, continue to operate it without oil may damage it.

9. Installation of tap-changer protection device and transmission device components

9.1 Installation of gas relay

Install according to the instruction manual provided when the gas relay is supplied.

9.2 Use of pressure relief device

The red bar on the top of the pressure relief device must be removed before operation (Figure 27).

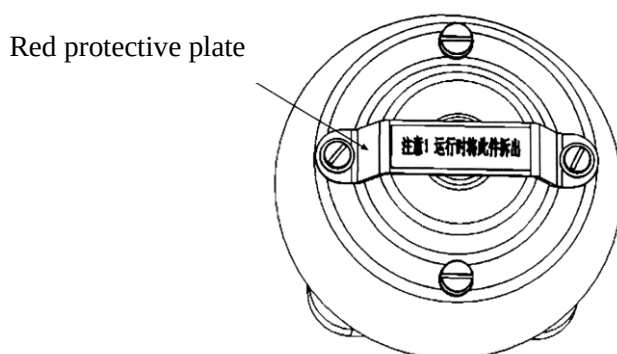


Figure 27 Pressure relief device

9.3 Installation of motor drive unit

Install according to the MDU instruction manual provided by HM.

9.4 Installation of bevel gear box

The bevel gear box is to be mounted onto the supporting bracket of the transformer tank by 2 bolts with 18mm bore diameter.



NOTE!

The serial number marked on the nameplate of the bevel gear box must be consistent with the serial number on the nameplate of the tap-changer.

When horizontal and vertical drive shafts are over two meters, an intermediate support gear box must be added. The installation of the specially designed bevel gear box, steering gear box and the intermediate support gear box of the vertical or horizontal drive shaft can be carried out according to the above instructions.

The gear box is fixed on the mounting flange with 3 pressing plates (each board with 2 bolts), and the angle can be adjusted according to user requirements.

- 1) Loosen the pressing plate of the gear box and pull out one end of the board slightly (6 pieces of M8 bolts, 13mm wrench). (Figure 28)
- 2) Turn the gear box to the required installation position by rotating the drive shaft of the gear box. (Figure 29)
- 3) Tighten the gear box pressure plate, and tighten the bolt (Torque 15 N.m.).



CAUTION!

1. The gear box mustn't rotate on its own.
2. When rotating the gear box, pay attention to avoid bumping the pressure relief valve (if existing.)

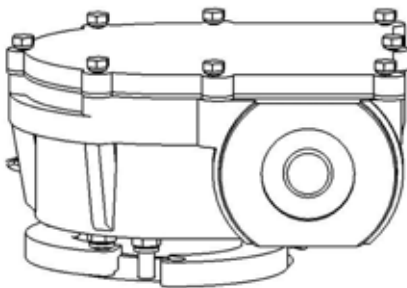


Figure 28 Gear box

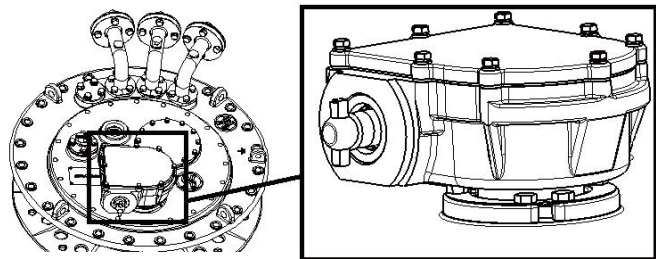


Figure 29 Turn the gear box

9.5 Installation of horizontal and vertical drive shafts

- 1) Install the horizontal and vertical drive shafts according to the relevant Huaming drive shaft instruction manual. (Figure 30)
- 2) Connect the drive shaft with motor drive unit and tap-changer according to the instruction manual of Huaming motor drive unit. (Figure 31)

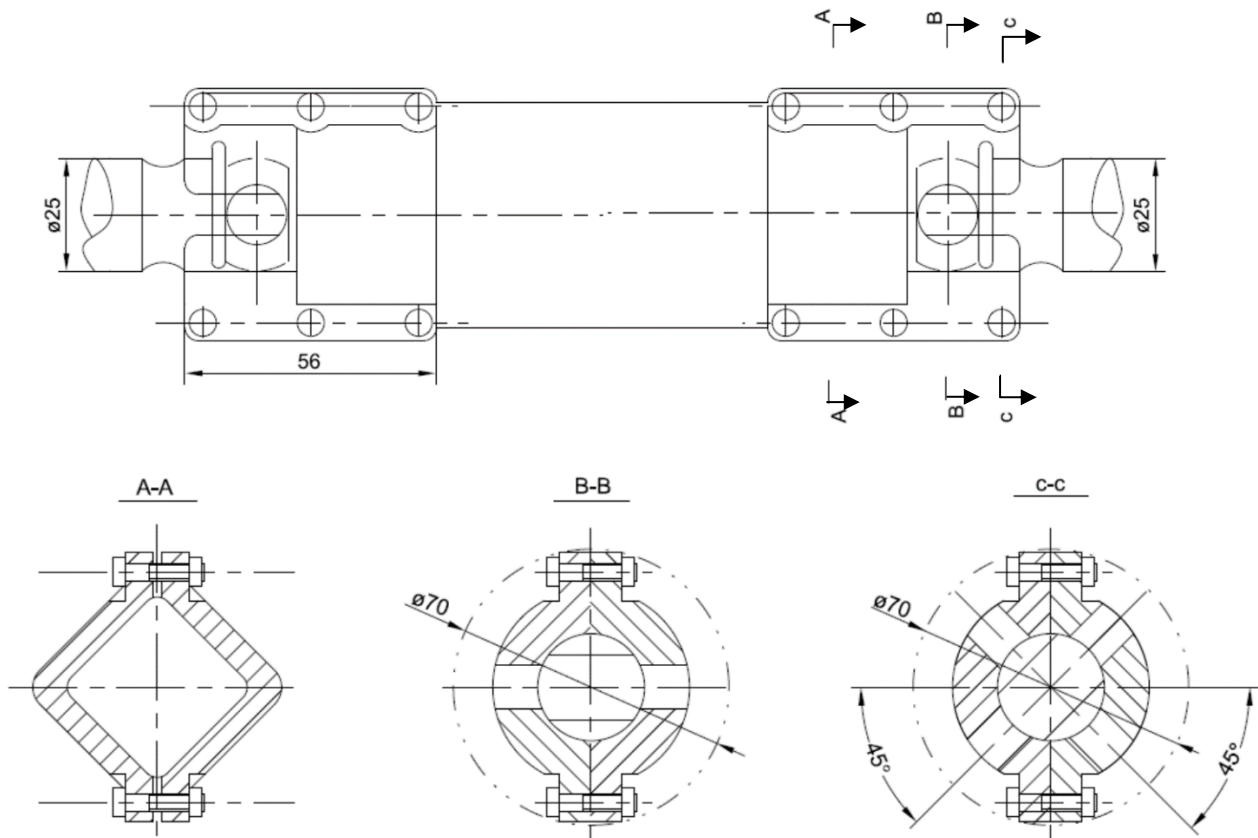


Figure 30 Install the drive shaft

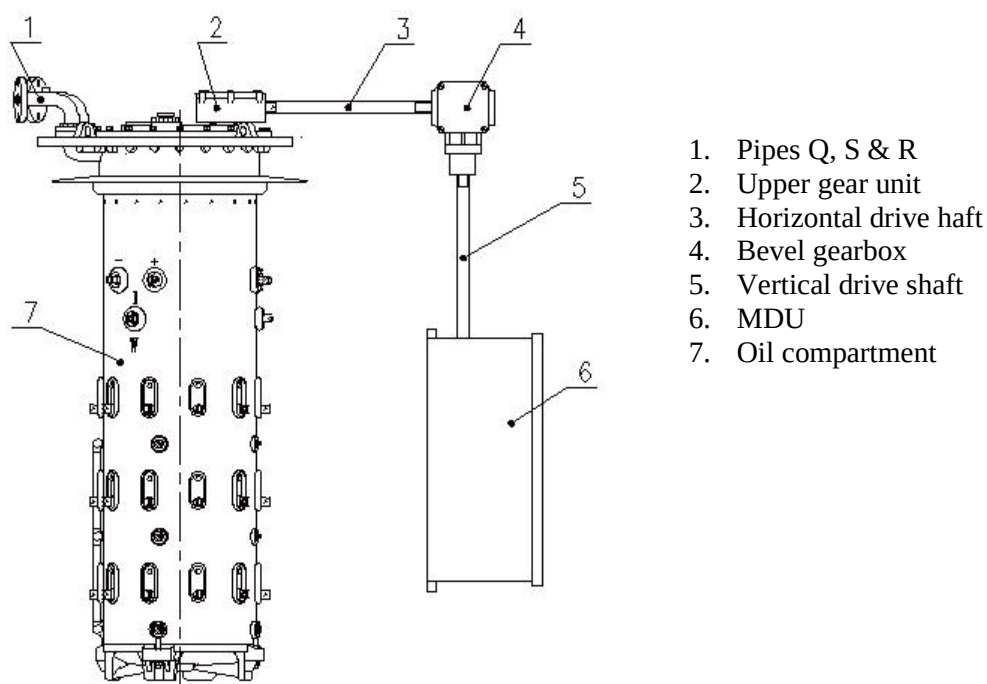


Figure 31 Drive shaft connection

**CAUTION!**

Incorrect adjustment of the gear box can cause damage to the tap-changer! The gear box can only be adjusted when the pressure ring is loosened. It must not rotate when it is adjusted. Only turn the drive shaft to adjust the gear box.

**NOTE!**

When the length of the vertical drive shaft exceeds 2m, a middle support box is required

9.6 Installation of driving shaft shield cover

9.6.1 Installation of horizontal shield cover

The length of the horizontal shield is determined by the design of the transformer manufacturer. After installation of the driving shaft, put the clamps on both ends of the horizontal shield covers (provided in the accessories), then clamp them on the protruding platform of the gear box and the bevel gear box, and finally move the clamps to the protruding platform positions on both sides, tighten it. (Figure 32)

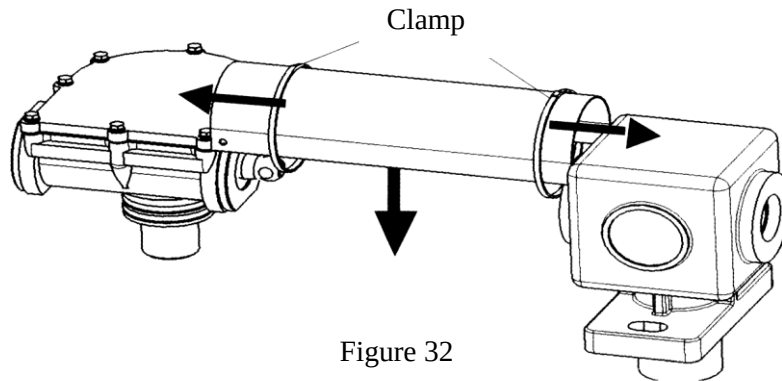


Figure 32

9.6.2 Installation of vertical shield cover

9.6.2.1 Fix 3 stainless steel 6×16 straight pins on the bushing (120° equally divided), and protrude the outer diameter of the bushing about 5mm, and the inner diameter positions should not be protruded. (Figure 33)

9.6.2.2 Place the bushing at the lower end of the bevel gear box and fix with 4 pieces of M6×10 set screws (90° equally divided). (Figure 34)

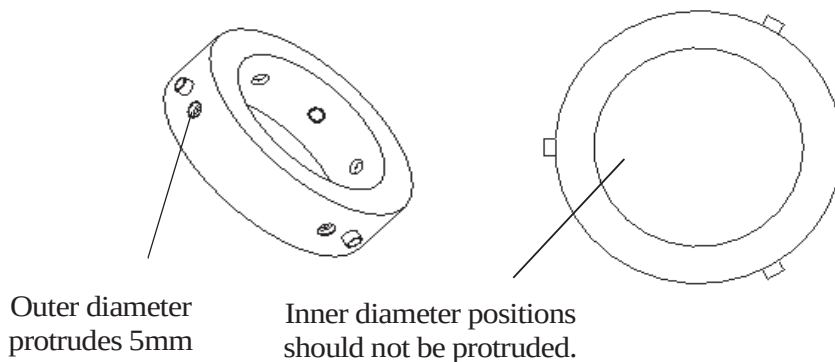


Figure 33 Fix steel

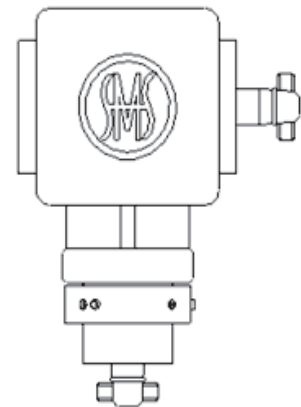


Figure 34 Place bushing

9.6.2.3 Put the two shield covers outside of the driving shaft with the gaps at both ends, and the thicker one on the top. Then connect it to the middle of the bevel gear box and motor drive unit according to the figures below. (Figures 35&36)

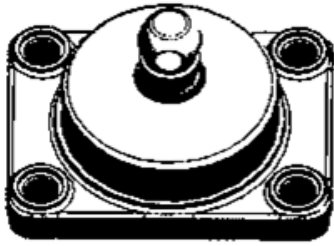


Figure 35 CMA7 connection to the drive shaft



Figure 36 Connect to the bevel gear box



Figure 37 Place the lower shield

9.6.2.4 Place the lower shield cover outside of the bushing gaps to avoid 4 bolts. (Figure 37)

9.6.2.5 Pull up the protective cover to the lower end of the square gear box, and set it on the bushing, with the 3 notches facing directly, 3 cylindrical pins, and then rotate the protective cover so that the cylindrical pins are just stuck in the horizontal groove (Figure 38). (3 pins are only for temporary hang)

9.6.2.6 At this time, the two protective covers have been positioned, use two clamps on the thinner protective cover at the bottom. Clamp the thicker protective cover on the upper part (Figure 39)

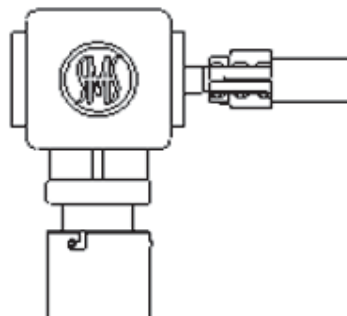
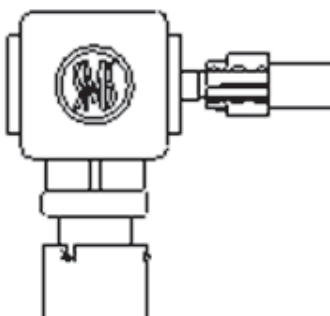


Figure 38 Rotate the protective cover



Figure 39 Clamp the protective cover

9.7 Installation of tap-changer group

The arrangement of tap-changer group composed of several units of tap-changers is quite special, and every tap-changer is at the setting position. The connection steps are as follows:

- 1) Check whether the operating positions of all on-load tap-changers are the same (check from the tap-changer head observation window and every tap-changer must be at the adjusting position).
- 2) Loosen the pressing ring of the gear box and slightly pull out one end of the ring (6 pieces of M8 bolts, 13mm wrench) (Figure 40).
- 3) Turn the gear box to the required installation position by rotating the drive shaft of the gear box.
- 4) Tighten the pressing ring of the gear box, and lock the bolt with a locking tab (Torque 15 N.m.).
- 5) Pay attention to the serial number printed on the flange of the gear box and the arrow below. The direction of the arrow indicates the steering direction of the drive shaft when using a hand crank to clockwise rotate the MDU. The direction of all arrows on all gear boxes must be the same. The gear box can be adjusted only when the pressing ring is loosened.

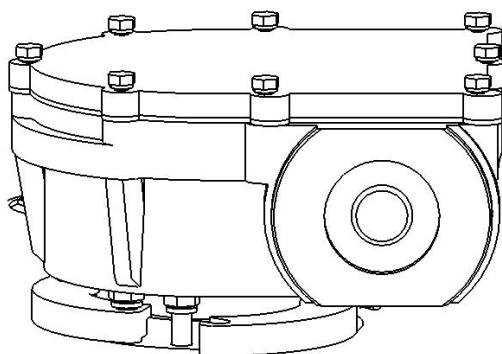


Figure 40 Loosen gearbox pressing ring



CAUTION!

The gear box can only be adjusted only by rotating its drive shaft. It must not rotate on its own.



CAUTION!

Incorrect adjustment of the gear box can cause damage to the tap-changer!

- 6) Rotate each shaft end counterclockwise to run every step of the tap-changer once, until the diverter switch operates for one full cycle.
- 7) Check whether the operating positions of every tap-changer head and the MDU are the same.
- 8) Install horizontal drive shafts between every tap-changer head. Each tap-changer has its own shaft coupling. Start with the tap-changer closest to the MDU.
- 9) After all the drive shafts are installed, the drive shaft of the gear box should continue to rotate for 2.5 cycles counterclockwise to correctly complete the tap change operation.
- 10) Rotate the drive shaft clockwise to adjust the tap-changer back to the calibration position. Then operate together with the diverter switch. The drive shaft of the gear box should continue to rotate for 2.5 cycles in the clockwise direction to make the tap change operation completed correctly.
- 11) Make sure that all tap-changers are operate (a clear sound can be heard) at the same time.
- 12) Check whether the operating positions of every tap-changer head and the MDU are the same.
- 13) Install the vertical drive shafts.

9.8 Tap position calibration of tap-changer transmission system

When the tap-changer is connected to the MDU, manual operation for one cycle before electric operation is required

When OLTC coupling with MDU, the time interval between the moment the diverter switch operates and the the completion of MDU operation is required to be the same for the two rotation directions.

The connection verification of the tap-changer and the MDU normally is performed before FAT test. But in order to ensure the reliability of the tap-changer, below procedure shall be applied for the connection verification.

- 1) Rotate in the 1→N direction. When the diverter switch operates (a clear sound will be heard), continue to rotate the hand crank and record the number of rotations, until the red mark in the green band on the MDU tap change operation indicator wheel appears in the middle of the observation window. Stop rotating and write down the number of rotations as “m”.
- 2) Rotate the hand crank in the opposite direction N→1 to return to the original setting position, and record the number of rotations as “K” (follow instruction step a)
- 3) If the number of rotations $m=k$, the connection is correct. If $m \neq k$ and $m-k > 1$, balance of the rotation difference is required. Loosen the connection of MDU's vertical shaft, rotate $(m-k) / 2$ laps in the direction which has more rotations, and finally connect the vertical drive shaft with the MDU.
- 4) Following the above steps, check the rotation difference between the MDU and the tap-changer, until the correct number of rotations, that is, $m=k$.

For example:

The connection verification between the CV tap-changer and the SHM MDU: from position 10 (setting position) to position 11, $m=5$ rotations; from position 11 back to 10 (the original setting position), $k=3$ rotations. The hand crank rotation difference: $m-k=5-3=2$. Adjust the number of rotations $(m-k) / 2 = (5-3) / 2 = 1$ rotation. Loosen the connection between the vertical drive shaft and the MDU, and turn the hand crank for one circle in the direction of position 10 to 11 according to the number of rotations calculated above. Then connect again. Check whether the rotation difference between the two directions is balanced:

- 1) Record the rotation numbers of m and k when connected
- 2) Disconnect the connection and rotate for $(m-k)/2$ circles in the direction with more rotations
- 3) Reconnect and verify until $m=k$

10. Tap-changer Factory test and test preparation

10.1 Test preparation

10.1.1 OLTC Fully gas release

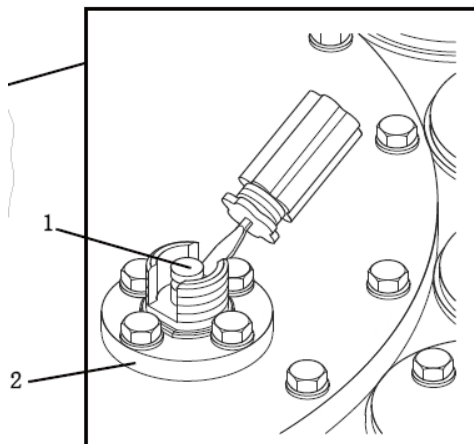
Before putting into operation for the first time, use the gas discharge stopper in the oil suction pipe on the tap-changer head to deflate the connection pipe.

10.1.1.1 Gas release

- 1) Remove the M30 bolt cap on the bleeding device E1 (36mm wrench) (Figure 41).
- 2) Pry up the valve stem with a screwdriver to deflate the tap-changer head (Figure 41)
- 3) Tighten the bolt cap on the bleeding device E1 (Torque 10 N.m.)

10.1.1.2 Gas release for S: connecting flange for suction pipe

- 1) Remove the M16 bolt cap on the pipe joint S (16mm wrench) (Figure 42).
- 2) Open the notched gas discharge stopper M6 to release gas.
- 3) Close the stopper (Torque 2 N.m.).
- 4) Close the bolt cap on the stopper. (Torque 9 N.m.)



1. Valve stem
2. Bleeding device E1

Figure 41 Pry up the valve stem

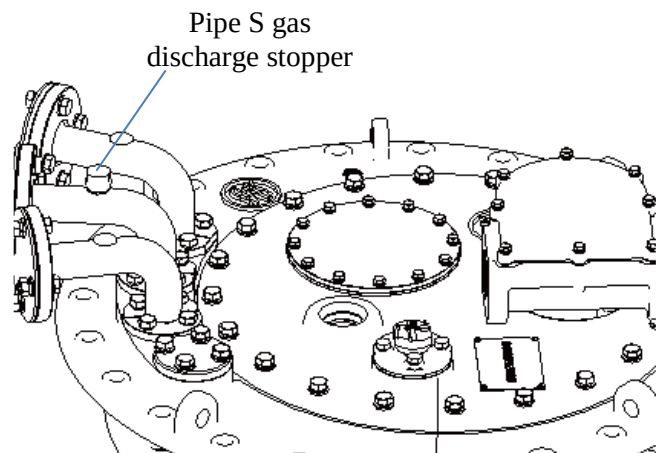


Figure 42 Remove the bolt cap

10.1.2 Grounding

- 1) Connect the grounding bolt of the tap-changer head to the transformer tank cover (1 piece of M12 bolt and nut) with a 19mm wrench, torque of 50 to 60 N.m. (Figure 43).
- 2) Connect the transmission mechanism's grounding bolt M12 (19mm wrench, torque 50 to 60 N.m.) connected to the transformer tank.

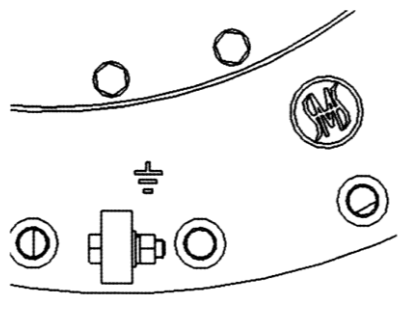


Figure 43 Connect the grounding bolt

10.2 Test in transformer factory

10.2.1 Operation test

Pre- operation must be carried out to check the mechanical functions of the tap-changer and the motor-drive unit before the transformer is energized.

**CAUTION!**

Continue to operate the tap-changer without oil will cause damage to the tap-changer !

When operating the tap-changer for the first time after the transformer is dried, make sure:

- 1) The tap selector is completely immersed in transformer oil and the diverter switch oil compartment is filled with oil
- 2) During the operation test, the tap-changer must run the complete operating range.

**NOTE!**

Make sure MDU's and tap-changer's position indicators show the same position.

**CAUTION!**

If the connection between the tap-changer and the MDU is improperly aligned, continuous operation will cause damage to the tap-changer.

- 3) Make sure that the MDU's and tap-changer's position indicators (observation window on the tap-changer head) show the same position in every operating position.
- 4) Check the electrical and mechanical end limit functions at two end positions (Please refer to MDU instruction manual).

10.2.2 The electrical test of the transformer

**DANGER!**

There is explosive gas under the tap-changer cover, in the connection system, oil conservator and the outlet of air dryer. The flying parts and the high temperature oil splashing to the outside will cause fatal and serious injury.

**NOTE!**

Please confirm: there is no open flame, heat source or electric spark (such as electric spark produced by static electricity) within its direct contact range, and there is no possibility of generating them. It must be ensured that the tap-changer test must be carried out by professionals in accordance with the test procedures.

11. Transformer transportation and put into operation on site

11.1 Remove the MDU for transportation

- 1) Place the MDU at the calibration position and then remove the coupling
- 2) Remove the MDU
- 3) Do not operate the MDU when the tap-changer is not connected well. The re-assembly of the MDU and drive shaft must be carried out according to Chapter 9.3 to 9.7 of this instruction manual.

**NOTE!**

Do not operate the MDU when the tap-changer is not well connected.

11.2 Oil filling and transportation without oil conservator

If the transformer is filled with oil during storage and transportation but no oil conservator is installed, a bypass pipe must be installed between the inside of the diverter switch oil chamber and the transformer oil tank to balance the pressure generated by oil expansion. The bypass tube is installed between the pipe connector E2 and Q.

**NOTE!**

For the storage period of 3 to 4 weeks without oil conservator, the oil level of the tap changer should be lowered by about 5 liters.

11.3 Transportation without oil

If the transformer is transported or stored without oil filling, all the oil in the tap-changer oil compartment must be discharged. The protection method for tap-changer oil compartment should be same as that for the transformer, such as nitrogen filling.

**NOTE!**

For long-term storage, the heater of the MDU should be regularly energized to prevent moisture.

11.4 Putting transformer into operation on site



DANGER!

There is explosive gas under the tap-changer cover, in the connection system, oil conservator and the outlet of air dryer. The flying parts and the high temperature oil splashing to the outside will cause fatal and serious injury.

Before putting the transformer into operation, make sure the tap selector is completely immersed in the transformer oil and the oil compartment is filled with oil.

- 1) Fill OLTC with oil according to Chapter 8.2 (Refer to Chapter 13.1 Table 6 for the limit of oil quality).
- 2) To fully release gas in OLTC, please follow Chapter 10.1.1
- 3) For pre-operation procedure, please follow Chapter 10.2.1
- 4) Please make sure that the OLTC oil conservator's lowest oil level signal contact is connected to the circuit breaker's trip circuit.
- 5) Check whether the gas relay function is normal according to the Huaming gas relay QJ-25 series instruction manual.
- 6) The protective strip of the pressure relief valve must be removed before putting it into operation. Follow safety instructions of all functional inspections and tests during commissioning and the safety instructions described in Chapters 9.2 and 10.2.2.



CAUTION!

Before OLTC operating, make sure that starting current instantaneous impact is fully subsided. Usually, the starting current is several times of the transformer rated current and therefore may cause overload to the OLTC during tap changing. After the transformer is energized and the starting current instantaneous impact is fully subsided, OLTC operation can be performed with or without load.



NOTE!

Please confirm: there is no open flame, heat source or electric spark (such as electric spark produced by static electricity) within its direct contact range, and there is no possibility of generating it.

Before the transformer is put into operation, please check the OLTC's and MDU's mechanical operation:

- 1) OLTC must do at least 10 complete cycles of switching.
- 2) Make sure the tap-changer's head cover position indicator and MDU are at the same position for every switching.
- 3) Check the electrical and mechanical end position limits at max. and min. positions.
- 4) Check the protective relay's functions. The protective relay must be connected to the tripping circuit of the circuit breakers. Make sure that the circuit breakers switch off the transformer when the protective relay works. Press the protective relay's test button "cutoff" and check if the circuit breaker cuts off the transformer's power. Only when the protective relay's "In Operation" button is pressed will the transformer be energized.
- 5) After energizing the transformer, the tap-changer test operation can be carried out on load. The accumulated gas beneath the OLTC head cover may cause small amount oil leakage.

12. Tap-changer operation monitoring and maintenance

**NOTE!**

It is necessary to regularly check the OLTC head cover, protective relay and MDU.

**CAUTION**

Inspection of OLTC head cover, protective relay and MDU should be carried out regularly.

- Whether there is leakage at the connection between the protection relay and the pipeline;
- The condition of the MDU sealing and all electric components;
- If the protective relay alarms, a thorough inspection should be done to the transformer and OLTC. If necessary, the tap-changer body can be lifted out for inspection.
- Before putting into operation again, OLTC and transformer must be inspected for damages.

**NOTE!**

Please contact Huaming service team if OLTC and MDU have severe problems or the protective relay actions and it's not easy to fix on site.

13. Maintenance

Danger!



Electric shock!

An energized transformer may cause death or serious injury.

- ▶ Turn off the high voltage and low voltage ends of the transformer.
- ▶ Lock the transformer to prevent accidental restart.
- ▶ Ensure that all components are powered off.
- ▶ Clearly ground all transformer terminals (ground wire, ground isolation switch) and short-circuit them.
- ▶ Cover or isolate adjacent live parts.

Danger!



Electric shock!

Operation of energized on-load tap-changers may cause personnel death or serious injury.

- ▶ Disconnect all auxiliary circuits, such as tap change monitoring devices, pressure relief valves and pressure monitors.
- ▶ Ensure that all components are powered off.

Warning!



Warning!

Risk of explosion!

Explosive gas in the oil compartment of the on-load tap-changer, transformer, piping system, oil conservator, and the outlet of the moisture absorber may ignite or explode, causing serious injury or death.

- ▶ Ensure that there are no open flames, hot surfaces, or electric sparks (for example, electric sparks caused by static electricity accumulation) in the environment around the transformer, and that the above situations will not occur.
- ▶ Do not operate any electrical equipment (for example, impact wrenches have the risk of sparking).
- ▶ Only use conductive and grounded hoses, pipes and pump operation equipment suitable for flammable liquids.

Remind

Check the motor drive unit!

Condensation in the MDU protection case can cause damage to MDU.

- ▶ Always keep MDU's protection case sealed.
- ▶ If the operation is interrupted for more than 2 weeks, connect and operate the anti-condensation heater in the MDU. If this requirement cannot be met (for example, the heater cannot be energized during transportation), a sufficient amount of desiccant must be placed inside the case.



NOTE!

OLTC must be inspected and maintained regularly to ensure its good performance. Otherwise, damages may be caused to OLTC and transformer operation.

13.1 Inspection

Please follow the inspection instruction below:

Follow the inspection instruction below:						
Interval	Instruction					
Once per year	Check the door tightness, cable bushings, MDU protective cabinet and control panel's ventilation device					
Once per year	Check the tightness of tap-changer head cover, protective relay and all pipes' connectors.					
Once per year	Check the MDU's protective cabinet and if the heater inside the control panel works normally or not.					
Once per year	Check if the protective relay functions normally.					
Once per year	Check if the Silica gel desiccant inside the OLTC oil conservator work normally					
Once per year	Check the oil sample. If it doesn't meet the parameters below, it's recommended to change the transformer oil.					
	OLTC equipment highest voltage (kV)	Oil before operation		Oil during operation		
		Water (mg/L)	Breakdown voltage (kV/2.5mm electrode)	Water (mg/L)	Breakdown voltage (kV/2.5mm electrode)	
		72.5	≤20	≥35	≤40	≥30
		40.5	≤20	≥30	≤40	≥30

Table 6 Inspection instruction

13.2 Maintenance interval

Interval	Instruction
After 2 years or 20,000 times OLTC operation (based on the number shown in MDU's counter)	Lift the switch insert for inspection. Regarding this, please contact Huaming's after-sales department. (This inspection is not mandatory.)
After the above inspection, every 7 years or when the operation times meet Table 9 (based on the number shown in MDU's counter)	Inspect the tap-changer. Regarding this, please contact Huaming's after-sales department.
After 800,000 times OLTC operation (based on the number shown in MDU's counter)	Change the spring energy system, insert and moving contact group. Regarding this, please contact Huaming's after-sales department.

Table 7 Maintenance interval

OLTC	Transformer rated current	OLTC operation times	
		Without online oil filter	With online oil filter
CVIII 350 Y/D CV I 350	$I \leq 100A$	100,000	150,000
	$100A < I \leq 200A$	70,000	150,000
	$200A < I \leq 350A$	70,000	150,000
CV I 700	$I \leq 350A$	70,000	140,000
	$350A < I \leq 700A$	70,000	120,000
SVIII 500 Y/D	$I \leq 350A$	70,000	140,000
	$350A < I \leq 500A$	50,000	120,000

Table 8 CV/SV OLTC Maintenance Interval

Please write to export@huaming.com for any need of support.

13.3 Recommendation of oil filter installation

- 1) In high humidity environment such as tropical or subtropical regions, to save the OLTC from extra moisture, online oil filter is recommended.
- 2) For frequent switching applications such as industrial transformer or daily operation over 50 times, HM strongly recommends online oil filter.

13.4 Notice



CAUTION

OLTC must be maintained strictly follow the manual to ensure its good performance. Otherwise, it will cause damages to operations of tap-changer and transformer.



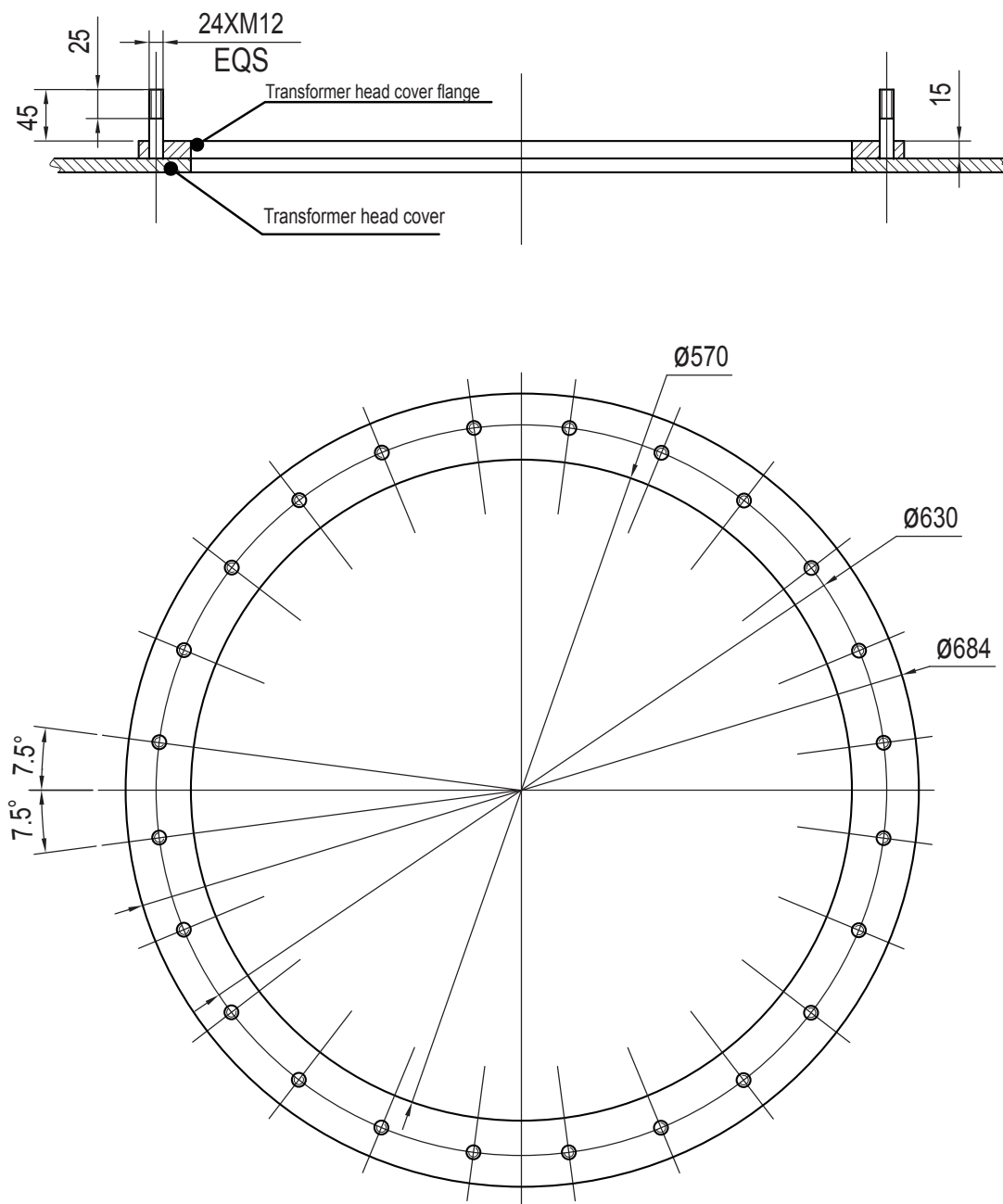
CAUTION

If the tap-changer does not regulate the voltage for a long time or has a tap position that is not used for a long time, no less than 5 cycles of tap change operation should be performed in the full tap range during a power cut.

14. Appendix

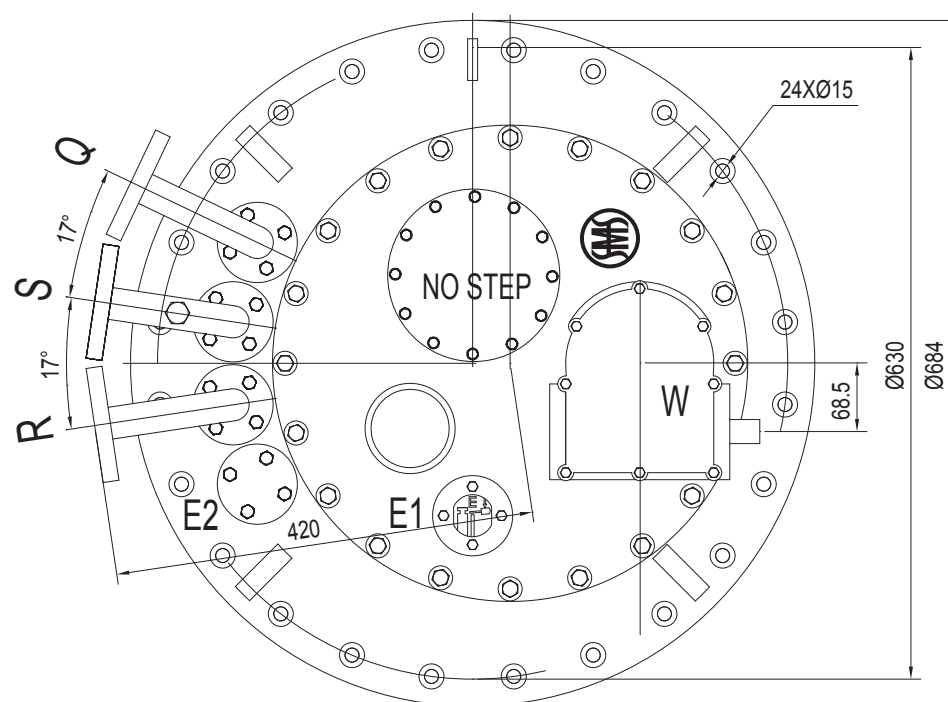
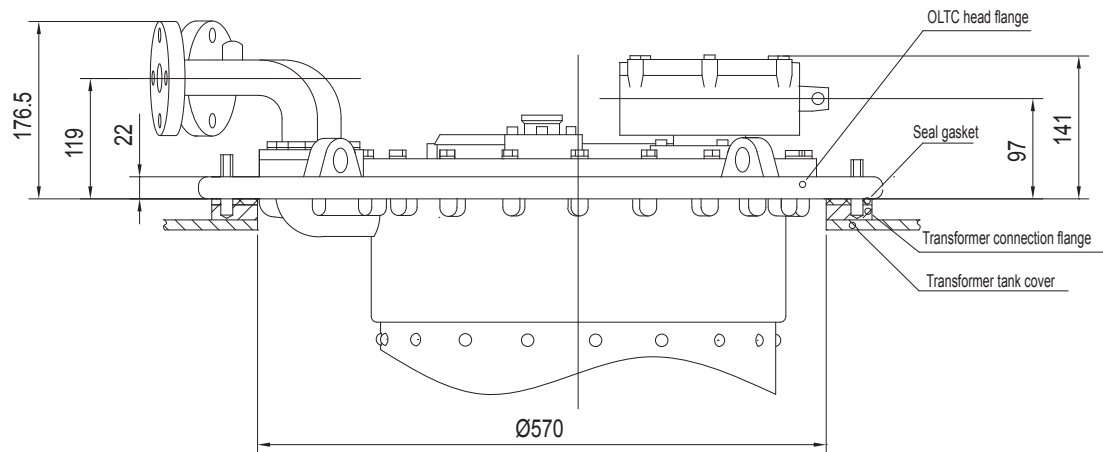
Appendix 1 Transformer connection flange, overall dimension	56
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Appendix 1 Transformer connection flange for CV OLTC, overall dimension

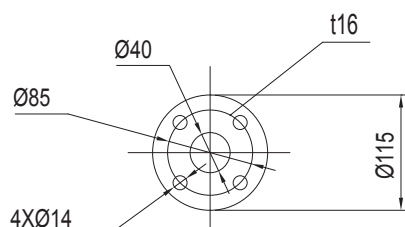


Unit: mm

Appendix 2 CV OLTC head flange for standard tank type, overall dimensions



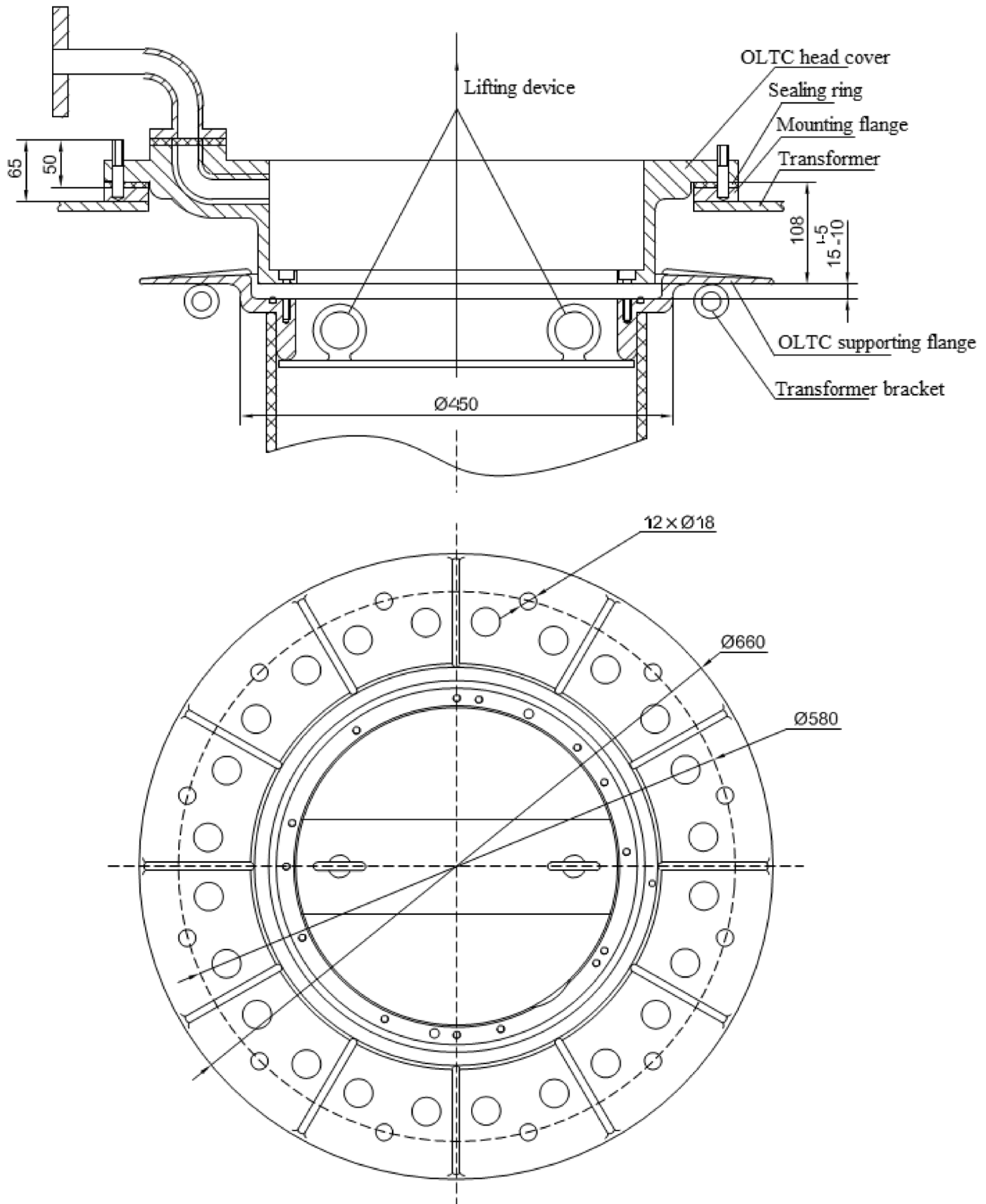
S/R/Q dimensions



E1: bleeding of tap changer head
E2: bleeding of transformer tank
R: connecting flange for protective relay
S: connecting flange for suction pipe
Q: connecting flange for oil return
W: head gear unit

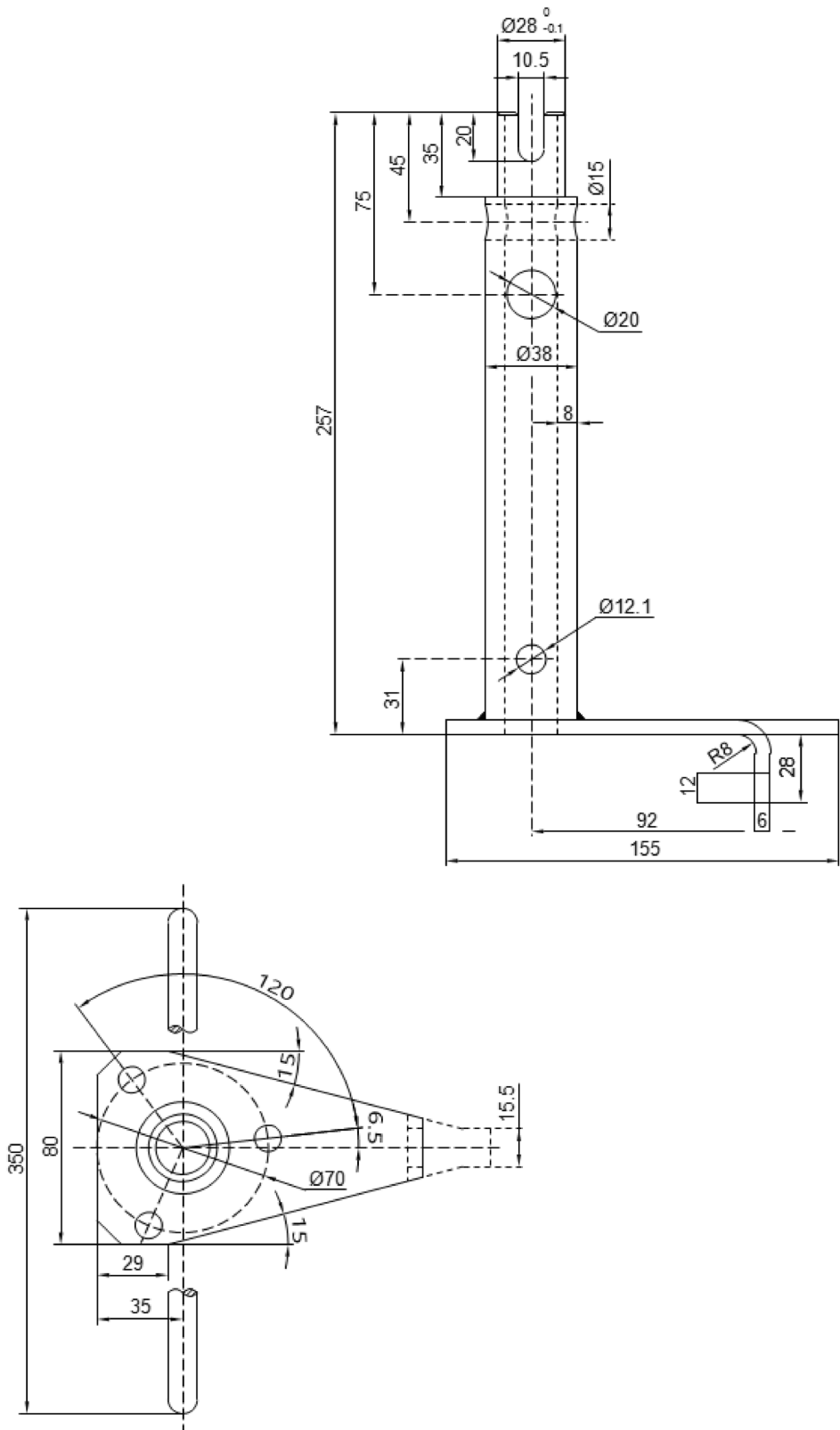
Unit: mm

Appendix 3 CV OLTC head flange installation for bell-type, overall dimensions



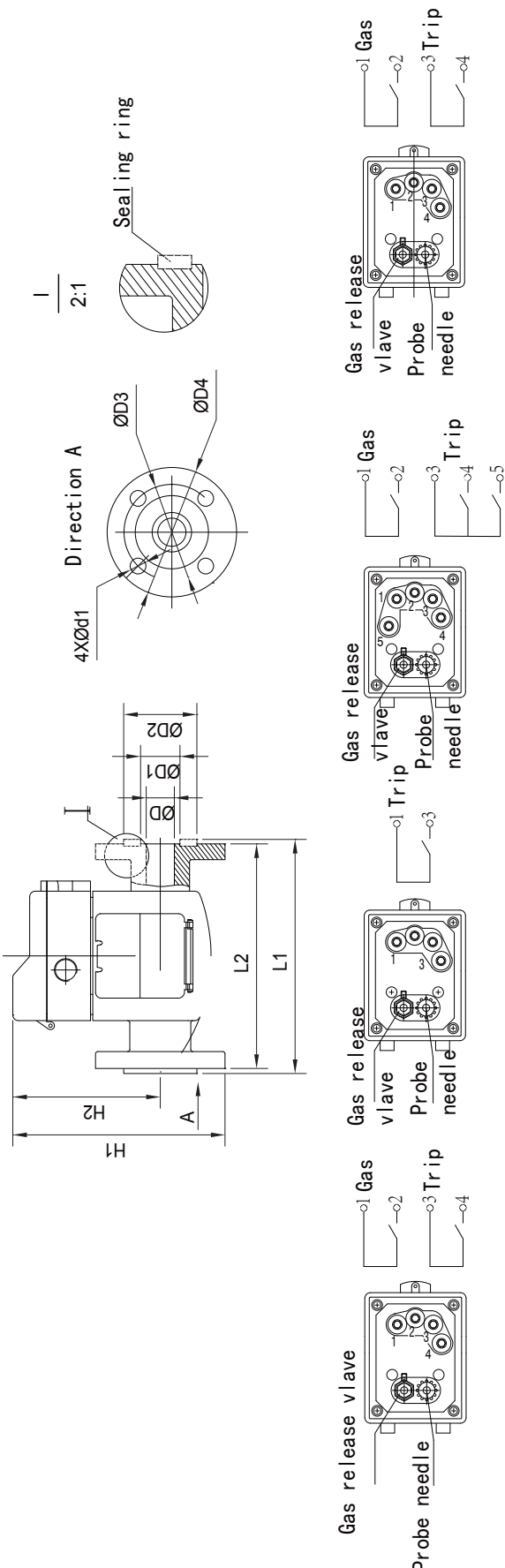
Unit: mm

Appendix 4 OLTC lifting tool diagram



Unit: mm

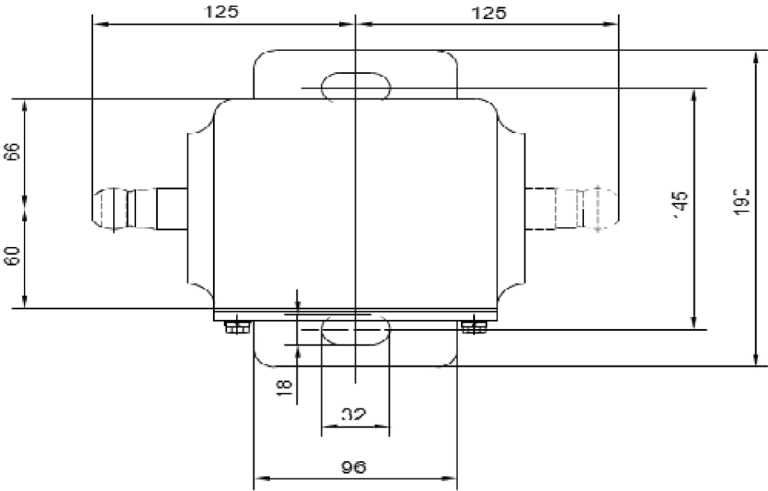
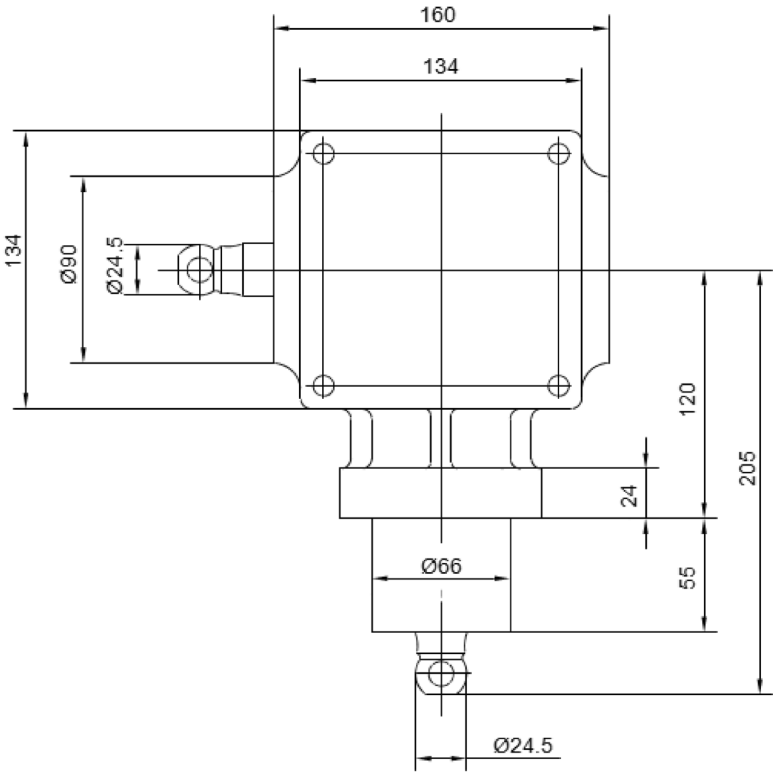
Appendix 5 Dimension and installation of protective relay



Remark: 1. Nylon cable fixing head MA20-10 (M20x1.5 external screw thread)
2. Probe needle is test button.

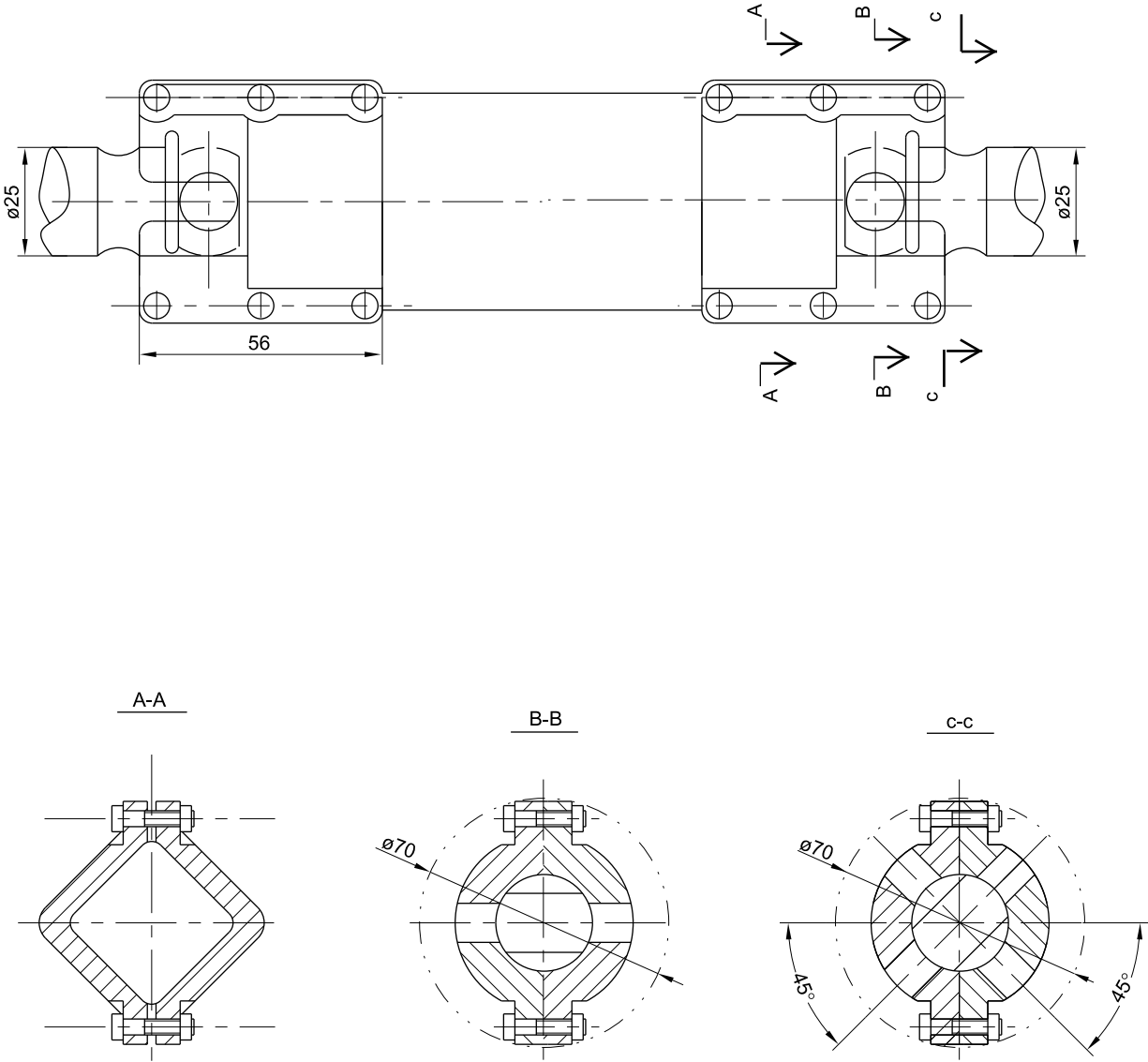
Model	D	D1	D2	D3	D4	d1	H 1	H 2	L1	L2	Remarks
QJ4-25	25	35	65	85	115	14	215	158	208	200	1 pair of gas signal and 1 pair of trip signal
QJ4G-25	25	35	65	85	115	14	190	133	208	200	1 pair of trip signal
QJ5-25	25	35	65	85	115	14	215	158	208	200	1 pair of gas signal, with common 2 pairs of trip signal
QJ6-25	25	35	65	85	115	14	190	133	208	200	2 independent trip signals
QJ7-25	25	35	65	85	115	14	215	158	208	200	1 gas signal, 2 independent trip signal

Appendix 6 Overall dimension of bevel gear



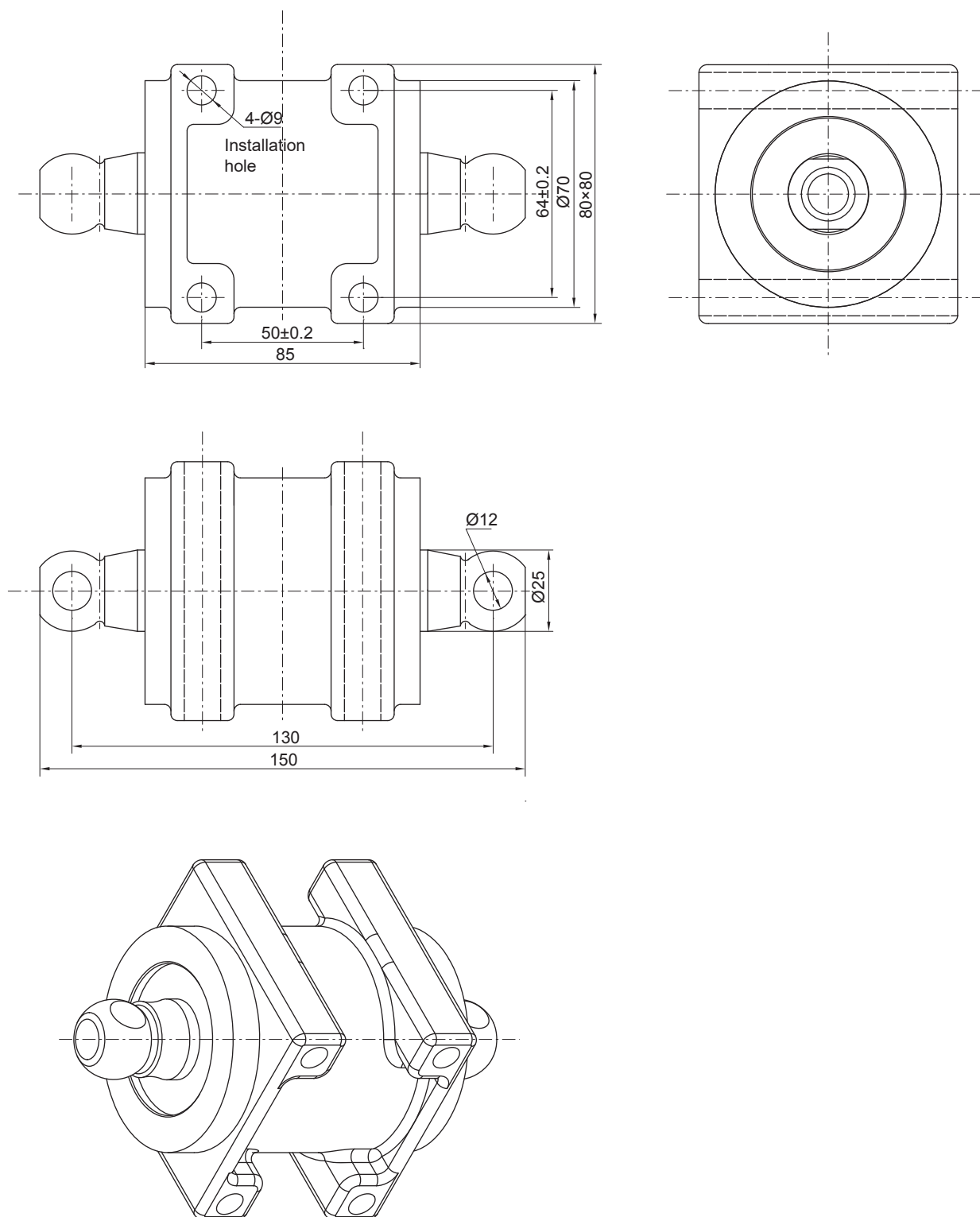
Unit: mm

Appendix 7 Installation of vertical and horizontal drive shafts



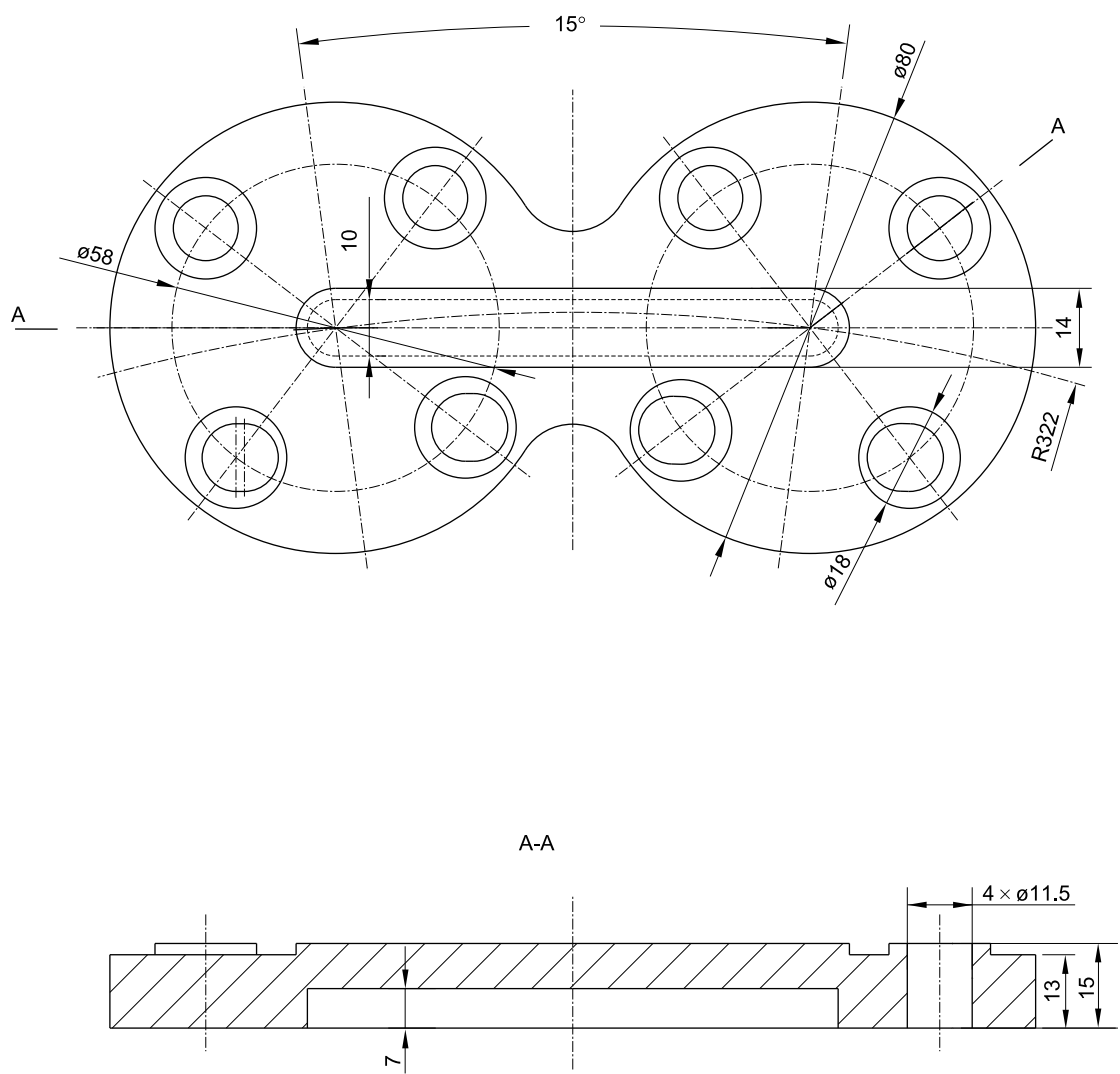
Unit: mm

Appendix 8 Dimension of middle support box



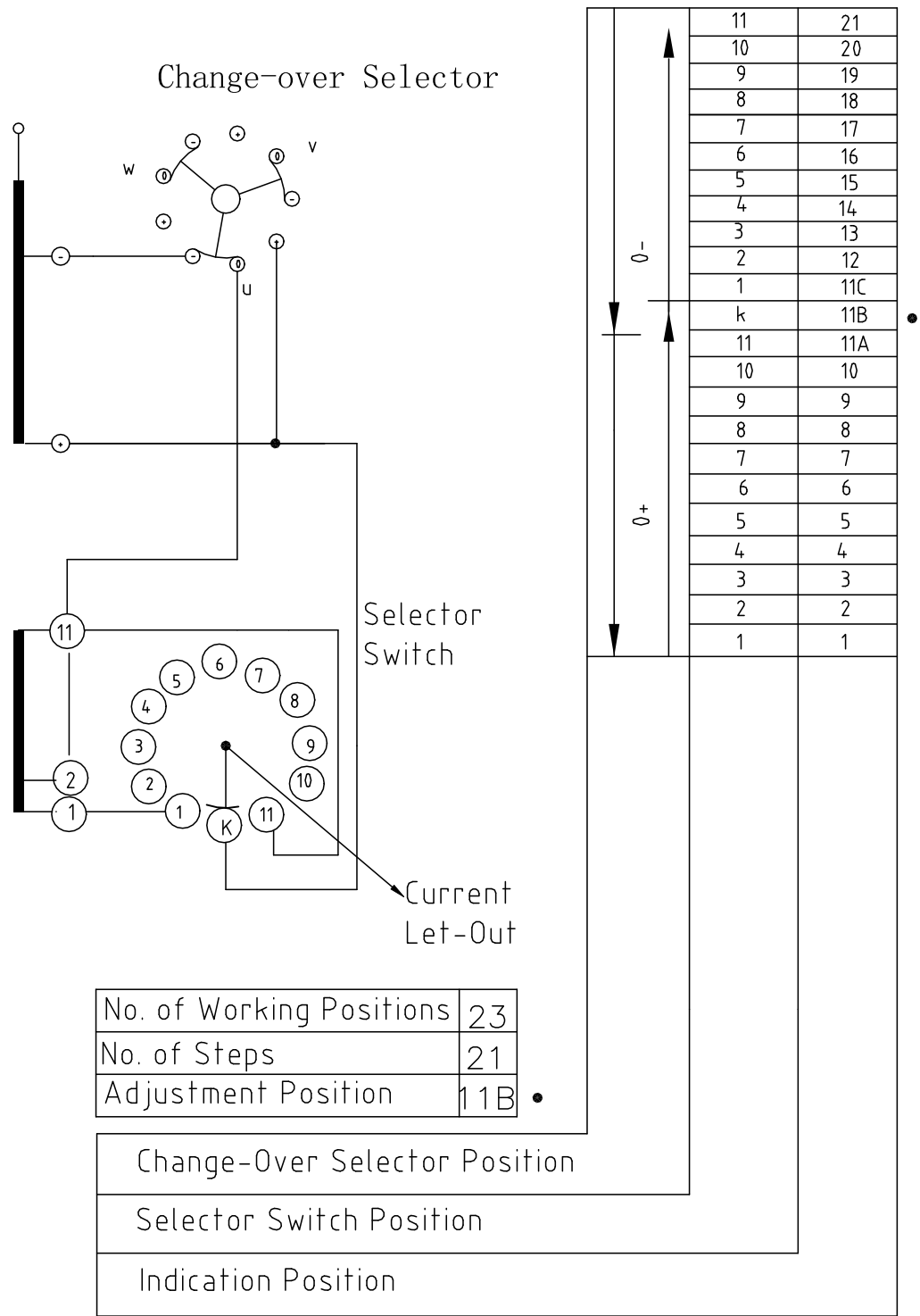
Unit: mm

Appendix 9 By-pass pipe structure



Unit: mm

Appendix 10 CV-III (12233G) operating position table and connection diagram





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